RANDOM DATA

ANALYSIS AND MEASUREMENT PROCEDURES

Second Edition
(Revised and Expanded)

JULIUS S. BENDAT
Mathematical Consultant in Random Data Analysis

ALLAN G. PIERSOL
Senior Scientist, Astron Research and Engineering

A Wiley-Interscience Publication
JOHN WILEY & SONS
New York · Chichester · Brisbane · Toronto · Singapore

CHAPTER 6

SINGLE-INPUT / OUTPUT RELATIONSHIPS

This chapter is concerned with the theory and applications of input/output relationships for single-input problems. It is assumed that records are from stationary random processes with zero mean values and that systems are constant-parameter linear systems. Single-input/single-output models and single-input/multiple-output models are discussed. Ordinary coherence functions are defined for these models. Multiple-input problems are covered in Chapter 7.

6.1 SINGLE-INPUT/SINGLE-OUTPUT MODELS

Consider a constant-parameter linear system with a weighting function $h(\tau)$ and frequency response function $H(f)$ as defined and discussed in Chapter 2. Assume the system is subjected to a well-defined single input $x(t)$ from a stationary random process $\{x(t)\}$ and produces a well-defined output $y(t)$, as illustrated in Figure 6.1. This output will belong to a stationary random process $\{y(t)\}$.

6.1.1 *Correlation and Spectral Relations*

Under ideal conditions, the output $y(t)$ for the system in Figure 6.1 is given by the convolution integral

$$y(t) = \int_0^\infty h(\tau)x(t-\tau)\,d\tau \tag{6.1}$$

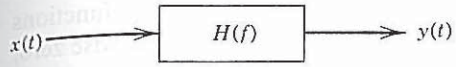


Figure 6.1 Ideal single-input/single-output system.

where $h(\tau) = 0$ for $\tau < 0$ when the system is physically realizable. The product $y(t)y(t+\tau)$ is given by

$$y(t)y(t+\tau) = \iint_0^\infty h(\alpha)h(\beta)x(t-\beta)x(t+\tau-\alpha)\,d\alpha\,d\beta \tag{6.2}$$

Taking expected values of both sides yields the *input/output autocorrelation relation*

$$R_{yy}(\tau) = \iint_0^\infty h(\alpha)h(\beta)R_{xx}(\tau+\beta-\alpha)\,d\alpha\,d\beta \tag{6.3}$$

Similarly, the product $x(t)y(t+\tau)$ is given by

$$x(t)y(t+\tau) = \int_0^\infty h(\alpha)x(t)x(t+\tau-\alpha)\,d\alpha \tag{6.4}$$

Here, expected values of both sides yields the *input/output cross-correlation relation*

$$R_{xy}(\tau) = \int_0^\infty h(\alpha)R_{xx}(\tau-\alpha)\,d\alpha \tag{6.5}$$

Note that Equation (6.5) is a convolution integral of the same form as Equation (6.1).

Direct Fourier transforms of Equations (6.3) and (6.5) after various algebraic steps yield two-sided spectral density functions $S_{xx}(f)$, $S_{yy}(f)$, and $S_{xy}(f)$, which satisfy the important formulas

$$S_{yy}(f) = |H(f)|^2 S_{xx}(f) \tag{6.6}$$

$$S_{xy}(f) = H(f)S_{xx}(f) \tag{6.7}$$

Here f may be either positive or negative. Note that Equation (6.6) is a real-valued relation containing only the gain factor $|H(f)|$ of the system. Equation (6.7) is a complex-valued relation, which can be broken down into a pair of equations to give both the gain factor $|H(f)|$ and the phase factor $\phi(f)$ of the system. Equation (6.6) is called the *input/output autospectrum relation*, while Equation (6.7) is called the *input/output cross-spectrum relation*. These results apply only to ideal situations where no extraneous noise exists at input or output points, and the systems have no time-varying or nonlinear characteristics. Interpretation of these spectral relations in the frequency domain is much easier than their corresponding correlation relations in the time domain.

In terms of physically measurable one-sided spectral density functions $G_{xx}(f)$, $G_{yy}(f)$, and $G_{xy}(f)$, where $G(f) = 2S(f)$ for $f \geq 0$, otherwise zero, Equations (6.6) and (6.7) become

$$G_{yy}(f) = |H(f)|^2 G_{xx}(f) \tag{6.8}$$

$$G_{xy}(f) = H(f) G_{xx}(f) \tag{6.9}$$

Let

$$G_{xy}(f) = |G_{xy}(f)| e^{-j\theta_{xy}(f)} \tag{6.10}$$

$$H(f) = |H(f)| e^{-j\phi(f)} \tag{6.11}$$

Then Equation (6.9) is equivalent to the pair of relations

$$|G_{xy}(f)| = |H(f)| G_{xx}(f) \tag{6.12}$$

$$\theta_{xy}(f) = \phi(f) \tag{6.13}$$

These results provide the basis for many engineering applications of spectral density functions. See Reference 6.1 for physical illustrations. Figure 6.2 shows how an input spectrum $G_{xx}(f)$ is modified in passing through a linear system described by $H(f)$.

From Equation (6.8), the output mean square value is given by

$$\psi_y^2 = \int_0^\infty G_{yy}(f)\, df = \int_0^\infty |H(f)|^2 G_{xx}(f)\, df \tag{6.14}$$

Equation (6.8) also permits the determination of $G_{xx}(f)$ from a knowledge of $G_{yy}(f)$ and $|H(f)|$, or the determination of $|H(f)|$ from a knowledge of $G_{xx}(f)$ and $G_{yy}(f)$. Equation (6.8) does not yield the compete frequency response function $H(f)$ of the system, however, since it contains no phase information. The complete frequency response function in both gain and phase can only be obtained from Equations (6.9) to (6.13) when both $G_{xy}(f)$ and $G_{xx}(f)$ are known.

An alternative direct transform is available to derive Equations (6.8) and (6.9) without first computing the correlation expressions of Equations (6.3) and (6.5). For any pair of long but finite records of length T, Equation (6.1) is equivalent to

$$Y(f) = H(f) X(f) \tag{6.15}$$

where $X(f)$ and $Y(f)$ are finite Fourier transforms of $x(t)$ and $y(t)$, respectively. It follows that

$$Y^*(f) = H^*(f) X^*(f)$$

$$|Y(f)|^2 = |H(f)|^2 |X(f)|^2$$

$$X^*(f) Y(f) = H(f) |X(f)|^2$$

Taking the expectation of the last two equations over different independent records, multiplying by $(2/T)$, and letting T increase without bound now proves from Equations (5.66) and (5.67) that

$$G_{yy}(f) = |H(f)|^2 G_{xx}(f) \tag{6.16}$$

$$G_{xy}(f) = H(f) G_{xx}(f) \tag{6.17}$$

Note the simplicity of this direct derivation. This method will be used in Section 6.1.4 and in Chapter 7.

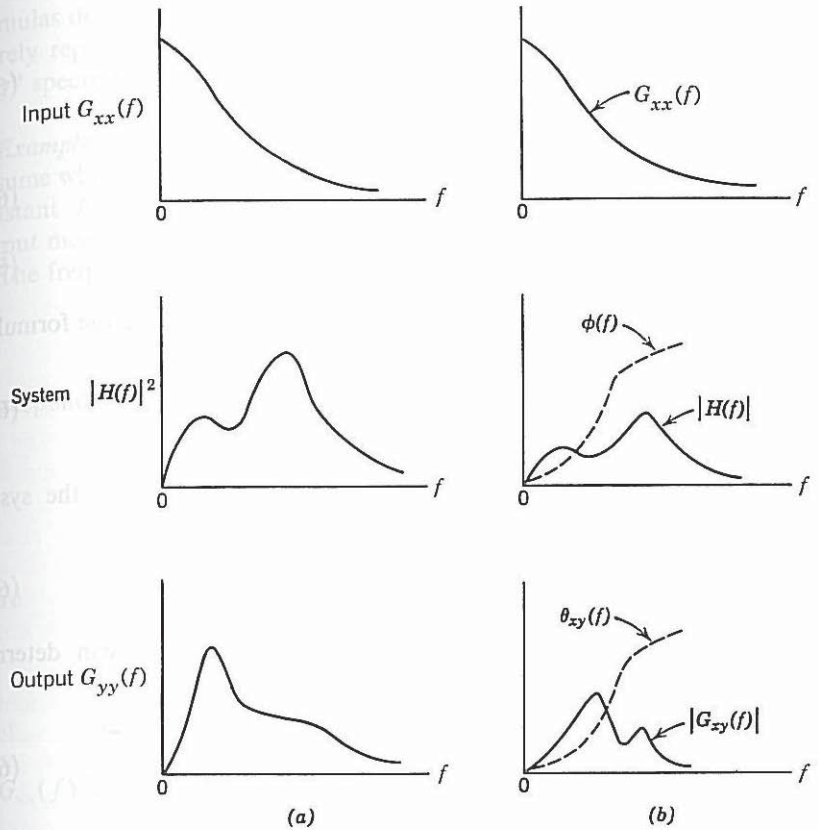


Figure 6.2 Input/output spectral relationships for linear systems. (*a*) Autospectra. (*b*) Cross-spectra.

Complex conjugation of Equation (6.17) yields the result

$$G_{xy}^{*}(f) = G_{yx}(f) = H^{*}(f)G_{xx}(f) \tag{6.18}$$

where

$$G_{yx}(f) = |G_{xy}(f)|e^{j\theta_{xy}(f)} \tag{6.19}$$

$$H^{*}(f) = |H(f)|e^{j\phi(f)} \tag{6.20}$$

Thus, to determine the phase factor of the system, one can use the formula

$$\frac{G_{xy}(f)}{G_{yx}(f)} = \frac{H(f)}{H^{*}(f)} = e^{-j2\phi(f)} \tag{6.21}$$

To determine the complete frequency response function of the system, Equations (6.16) and (6.18) show that

$$G_{yy}(f) = H(f)[H^{*}(f)G_{xx}(f)] = H(f)G_{yx}(f) \tag{6.22}$$

Hence, for the ideal single-input/single-output model, one can determine $H(f)$ using Equation (6.17) to yield

$$H(f) = \frac{G_{xy}(f)}{G_{xx}(f)} \tag{6.23}$$

while Equation (6.22) gives

$$H(f) = \frac{G_{yy}(f)}{G_{yx}(f)} \tag{6.24}$$

Thus

$$\frac{G_{xy}(f)}{G_{xx}(f)} = \frac{G_{yy}(f)}{G_{yx}(f)} \tag{6.25}$$

which is equivalent to

$$|G_{xy}(f)|^2 = G_{xx}(f)G_{yy}(f) \tag{6.26}$$

For transient data, discussed in Chapter 12, "energy" spectral density functions are used instead of "power"-type spectral density functions defined in Chapters 5 and 6. These are related by $\mathcal{G}_{xy}(f) = TG_{xy}(f)$, where $\mathcal{G}_{xy}(f)$ represents the "energy" cross-spectral density function. The transients $x(t)$ and $y(t)$ are assumed to exist only in the range $0 \le t \le T$. Input/output formulas derived in this chapter and Chapter 7 apply to such transient data by merely replacing "power" spectral density functions by corresponding "energy" spectral density functions.

Example 6.1. **Response Properties of Low-Pass Filter to White Noise.** Assume white noise is applied to the input of a low-pass RC filter with a time constant $K = RC$. Determine the output autospectral density function, the output mean square value, and the output autocorrelation function.

The frequency response function of the low-pass filter is

$$H(f) = (1 + j2\pi Kf)^{-1} = |H(f)|e^{-j\phi(f)}$$

corresponding to a weighting function of

$$h(\tau) = \begin{cases} \dfrac{1}{K}e^{-\tau/K} & \tau \ge 0 \\ 0 & \tau < 0 \end{cases}$$

Here

$$|H(f)| = \left[1 + (2\pi Kf)^2\right]^{-1/2}$$

$$\phi(f) = \tan^{-1}(2\pi Kf)$$

If $G_{xx}(f)$ is white noise where $G_{xx}(f) = A$, a constant for all $f \ge 0$, then

$$G_{yy}(f) = |H(f)|^2 G_{xx}(f) = \frac{A}{1 + (2\pi Kf)^2}$$

$$\psi_y^2 = \int_0^\infty G_{yy}(f)\,df = \int_0^\infty \frac{A}{1 + (2\pi Kf)^2}\,df = \frac{A}{4K}$$

$$R_{yy}(\tau) = \int_0^\infty G_{yy}(f)\cos 2\pi f\tau\,df = \frac{A}{4K}e^{-|\tau|/K}$$

Example 6.2. **Response Properties of Low-Pass Filter to Sine Wave.** Assume a sine wave with an autospectral density function

$$G_{xx}(f) = (X^2/2)\delta(f - f_0)$$

is applied to the low-pass RC filter in Example 6.1. Determine the output autospectral density function, the output mean square value, and the output autocorrelation function.

For this problem,

$$G_{yy}(f) = |H(f)|^2 G_{xx}(f) = \frac{(X^2/2)\delta(f - f_0)}{1 + (2\pi Kf)^2}$$

$$\psi_y^2 = \int_0^\infty G_{yy}(f)\, df = \frac{X^2/2}{1 + (2\pi Kf)^2}$$

$$R_{yy}(\tau) = \int_0^\infty G_{yy}(f)\cos 2\pi f\tau\, df = \frac{X^2/2}{1 + (2\pi Kf_0)^2}\cos 2\pi f_0\tau$$

Example 6.3. **Force-Input / Displacement-Output System.** Determine the output autospectral density function, the output autocorrelation function, and the output mean square value when the input is white noise, for the force-input/displacement-output system in Figure 2.2. These results apply also to other analogous systems, as discussed in Chapter 2.

Assume $G_{xx}(f) = G$, a constant. Then, from Equation (2.24a) or Table 2.1, when the force is expressed in displacement units, that is $x(t) = F(t)/k$, the output autospectral density function becomes

$$G_{yy}(f) = |H(f)|^2_{f-d} G = \frac{G}{\left[1 - (f/f_n)^2\right]^2 + (2\zeta f/f_n)^2} \qquad 0 \le f < \infty$$

The corresponding output autocorrelation function is given by

$$R_{yy}(\tau) = \frac{G\pi f_n e^{-2\pi f_n \zeta|\tau|}}{4\zeta}\left[\cos\left(2\pi f_n\sqrt{1-\zeta^2}\,|\tau|\right) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(2\pi f_n\sqrt{1-\zeta^2}\,|\tau|\right)\right]$$

The output mean square value is

$$\psi_y^2 = \int_0^\infty G_{yy}(f)\, df = R_{yy}(0) = \frac{G\pi f_n}{4\zeta}$$

Thus, for a noise input, the output mean square value ψ_y^2 is inversely proportional to ζ.

For a sine wave input, it will now be shown that the maximum value of ψ_y^2 is inversely proportional to ζ^2. Consider

$$F(t) = kx(t) = kX\cos 2\pi f_0 t \qquad f_0 = (1/T)$$

passing through a force-input/displacement-output system specified by the $H(f)$ in Figure 2.2. Here, the output $y(t)$ becomes

$$y(t) = kX|H(f_0)|\cos[2\pi f_0 t - \phi(f_0)]$$

with

$$\psi_y^2 = \frac{1}{T}\int_0^T y^2(t)\, dt = k^2(X^2/2)|H(f_0)|^2$$

From Equation (2.26), for small ζ,

$$\max|H(f_0)| = |H(f_r)| \approx \frac{1}{2k\zeta}$$

Hence

$$\max \psi_y^2 \approx \frac{X^2}{8\zeta^2}$$

proving the stated result.

Example 6.4. **Displacement-Input / Displacement-Output System.** Determine the output autospectral density function, the output autocorrelation function, and the output mean square value when the input is white noise, for the displacement-input/displacement-output system in Figure 2.4. These results apply also to other analogous systems, as discussed in Chapter 2.

Assume $G_x(f) = G$, a constant. Then, from Equation (2.38a) or Table 2.1, the output autospectral density function becomes

$$G_{yy}(f) = |H(f)|^2_{d-d} G = \frac{G\left[1 + (2\zeta f/f_n)^2\right]}{\left[1 - (f/f_n)^2\right]^2 + (2\zeta f/f_n)^2}$$

The corresponding output autocorrelation function is given by

$$R_{yy}(\tau) = \frac{G\pi f_n(1 + 4\zeta^2)}{4\zeta} e^{-2f_n\zeta|\tau|}\left[\cos\left(2\pi f_n\sqrt{1-\zeta^2}\,|\tau|\right) + \frac{\zeta(1 - 4\zeta^2)}{\sqrt{1-\zeta^2}(1 + 4\zeta^2)}\sin\left(2\pi f_n\sqrt{1-\zeta^2}\,|\tau|\right)\right]$$

The output mean square value is

$$\psi_y^2 = \int_0^\infty G_{yy}(f)\, df = R_{yy}(0) = \frac{G\pi f_n(1 + 4\zeta^2)}{4\zeta}$$

The importance of exponential-cosine and exponential-sine autocorrelation functions for many physical problems is apparent from the last two examples. In cases where $\zeta \ll 1$, results can be approximated using only exponential-cosine functions.

6.1.2 *Ordinary Coherence Functions*

Assuming $G_{xx}(f)$ and $G_{yy}(f)$ are both different from zero and do not contain delta functions, the coherence function between the input $x(t)$ and the output $y(t)$ is a real-valued quantity defined by

$$\gamma_{xy}^2(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)G_{yy}(f)} = \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)} \tag{6.27}$$

where the G's are the one-sided spectra and the S's are the two-sided theoretical spectra defined previously. From Equation (5.85), it follows that the coherence function satisfies for all f

$$0 \le \gamma_{xy}^2(f) \le 1 \tag{6.28}$$

To eliminate delta functions at the origin, mean values different from zero should be removed from the data before applying these last two results. Note that the coherence function is analogous to the square of the correlation function coefficient $\rho_{xy}^2(\tau)$ defined by Equation (5.16).

For a constant-parameter linear system, Equations (6.16) and (6.17) apply and may be substituted into Equations (6.27) to obtain

$$\gamma_{xy}^2(f) = \frac{|H(f)|^2 G_{xx}^2(f)}{G_{xx}(f)|H(f)|^2 G_{xx}(f)} = 1 \tag{6.29}$$

Hence for the ideal case of a constant-parameter linear system with a single clearly defined input and output, the coherence function will be unity. If $x(t)$ and $y(t)$ are completely unrelated, the coherence function will be zero. If the coherence function is greater than zero but less than unity, one or more of three possible physical situations exist.

a. Extraneous noise is present in the measurements.
b. The system relating $x(t)$ and $y(t)$ is not linear.
c. $y(t)$ is an output due to an input $x(t)$ as well as to other inputs.

For linear systems, the coherence function $\gamma_{xy}^2(f)$ can be interpreted as the fractional portion of the mean square value at the output $y(t)$ that is contributed by the input $x(t)$ at frequency f. Conversely, the quantity $[1 - \gamma_{xy}^2(f)]$ is a measure of the mean square value of $y(t)$ not accounted for by $x(t)$ at frequency f.

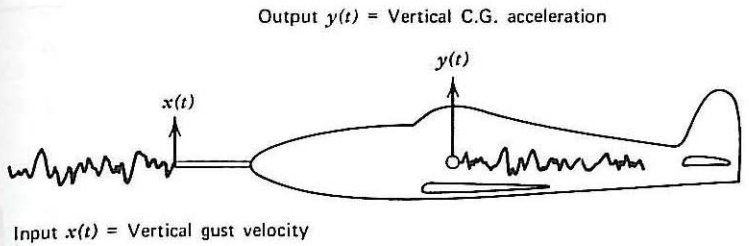


Figure 6.3 Airplane flying through atmospheric turbulence.

***Example 6.5.* Physical Illustration of Coherence Measurement.** Consider an airplane flying through a patch of atmospheric turbulence, as illustrated in Figure 6.3. Let the input $x(t)$ be vertical gust velocity in feet/second as measured with a probe extending forward of the airplane, and the output $y(t)$ be vertical acceleration in g's measured with an accelerometer at the center of gravity of the airplane. The resulting coherence function and autospectra for actual data of this type are presented in Figures 6.4 and 6.5. In this problem, the spectral data were computed over a frequency range from 0.1 to 4.0 Hz with a resolution bandwidth of 0.05 Hz and a record length of about 10 min.

From Figure 6.4, it is seen that the input gust velocity and output airplane acceleration display a relatively strong coherence of 0.8 to 0.9 over the frequency range from about 0.4 to about 2.0 Hz. Below and above this range, however, the coherence function diminishes. At the lower frequencies, the vertical acceleration of the airplane is increasingly due to maneuver loads induced through the control system by the pilot, rather than due to atmospheric turbulence loads. Hence, the loss of coherence at the lower frequencies probably reflects contributions to the output $y(t)$ from inputs other than the measured input $x(t)$. At the higher frequencies, the low pass filtering characteristics of the airplane response plus the decaying nature of the input autospectrum cause the output autospectrum to fall off sharply, as indicated in Figure 6.5. On the other hand, the noise floor for the data acquisition and recording equipment generally does not fall off with increasing frequency. Hence, the diminishing coherence at the higher frequencies probably results from the contributions of extraneous measurement noise. This concludes Example 6.5.

For any two arbitrary records $x(t)$ and $y(t)$, one can always compute the ordinary coherence function from $G_{xx}(f)$, $G_{yy}(f)$, and $G_{xy}(f)$. The value of this coherence function indicates how much of one record is linearly related to the other record. It does not necessarily indicate a cause-and-effect relationship between the two records. The issue of causality is addressed in Section 6.2.1.

For applications of coherence functions to problems of estimating linear frequency response functions, the coherence function may be considered to be the ratio of two different measures of the square of the system gain factor.

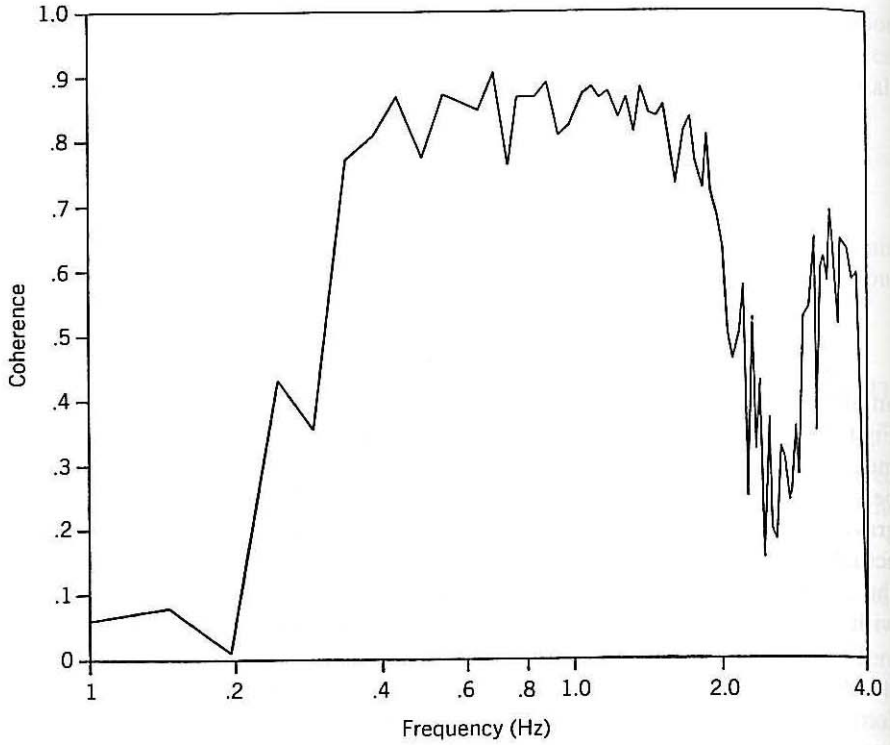


Figure 6.4 Coherence function between gust velocity and response acceleration. These data resulted from studies funded by the NASA Langley Research Center, Hampton, Virginia, under Contract NAS 1-8538.

From Equation (6.16), one measure is given by

$$|H(f)|_1^2 = \frac{G_{yy}(f)}{G_{xx}(f)} \tag{6.30}$$

From Equation (6.17), the second measure is given by

$$|H(f)|_2^2 = \frac{|G_{xy}(f)|^2}{G_{xx}^2(f)} \tag{6.31}$$

Now, their ratio gives the coherence function

$$\frac{|H(f)|_2^2}{|H(f)|_1^2} = \frac{|G_{xy}(f)|^2}{G_{xx}(f)G_{yy}(f)} = \gamma_{xy}^2(f) \tag{6.32}$$

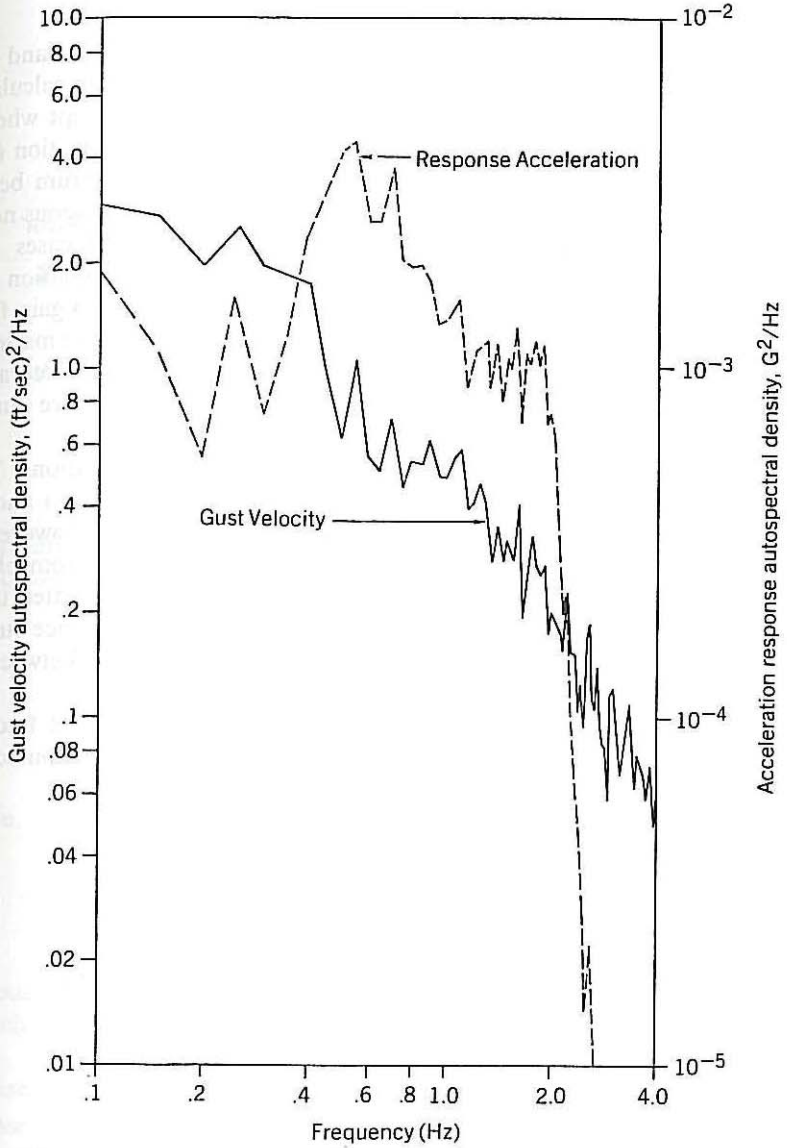


Figure 6.5 Autospectra of gust velocity and response acceleration. These data resulted from studies funded by the NASA Langley Research Center, Hampton, Virginia, under Contract NAS 1-8538.

In practice, measured values of Equation (6.32) will be between zero and unity. The gain factor estimate of Equation (6.30), based on autospectra calculations of input and output, will be a biased estimate for all cases except when the coherence function equals unity. The gain factor estimate of Equation (6.31), however, based on the input autospectrum and the cross-spectrum between input and output, will be a biased estimate for cases where extraneous noise is present at the input, but will be an unbiased estimate for cases where extraneous noise is present at the output only. In particular, Equation (6.31) provides an unbiased estimate of the frequency response function gain factors in multiple-input problems when the inputs are uncorrelated. These matters are discussed further in Chapter 9, where it is shown how the accuracy of frequency response function estimates increases as the coherence function approaches unity.

Coherence functions are preserved under linear transformations. To be specific, suppose one desires the coherence function between $x(t)$ and $y(t)$ where these two quantities cannot be measured easily. Assume, however, that one can measure two other quantities $x_1(t)$ and $y_1(t)$, where, from physical considerations, it can be stated that $x_1(t)$ is perfectly linearly related to $x(t)$ and $y_1(t)$ is perfectly linearly related to $y(t)$. Then the coherence function between $x_1(t)$ and $y_1(t)$ will give the desired coherence function between $x(t)$ and $y(t)$. The proof is as follows.

Perfect linear relationships mean that there exist hypothetical frequency response functions $A(f)$ and $B(f)$, which do not have to be computed, such that

$$X_1(f) = A(f)X(f) \qquad Y_1(f) = B(f)Y(f)$$

Then, at any value of f,

$$G_{x_1x_1} = |A|^2 G_{xx} \qquad G_{y_1y_1} = |B|^2 G_{yy} \qquad G_{x_1x_1} = A^*BG_{xy}$$

Hence,

$$\gamma^2_{x_1y_1} = \frac{|G_{x_1y_1}|^2}{G_{x_1x_1}G_{y_1y_1}} = \frac{|A^*B|^2|G_{xy}|^2}{|A|^2G_{xx}|B|^2G_y} = \frac{|G_{xy}|^2}{G_{xx}G_{yy}} = \gamma^2_{xy}$$

This result is important for many applications discussed in Reference 6.1.

6.1.3 *Models with Extraneous Noise*

Consider models where extraneous noise is measured at the input and output points to a linear system $H(f)$. Let the true signals be $u(t)$ and $v(t)$ and the extraneous noise be $m(t)$ and $n(t)$, respectively, as shown in Figure 6.6. Assume that only $u(t)$ passes through the system to produce the true output

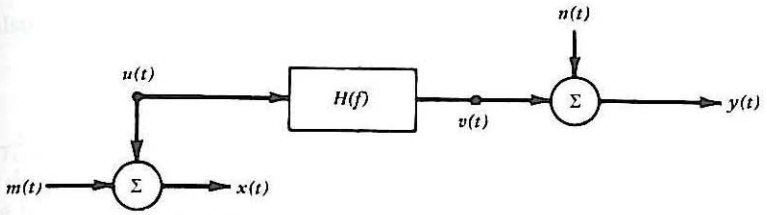


Figure 6.6 Single-input/single-output system with extraneous noise.

$v(t)$, but that the measured input and output records are

$$\begin{aligned} x(t) &= u(t) + m(t) \\ y(t) &= v(t) + n(t) \end{aligned} \tag{6.33}$$

For arbitrary correlations between the signal and noise terms, autospectral and cross-spectral density functions for $x(t)$ and $y(t)$ will be

$$\begin{aligned} G_{xx}(f) &= G_{uu}(f) + G_{nn}(f) + G_{um}(f) + G_{mu}(f) \\ G_{yy}(f) &= G_{vv}(f) + G_{nn}(f) + G_{vn}(f) + G_{nv}(f) \\ G_{xy}(f) &= G_{uv}(f) + G_{un}(f) + G_{mv}(f) + G_{mn}(f) \end{aligned} \tag{6.34}$$

where

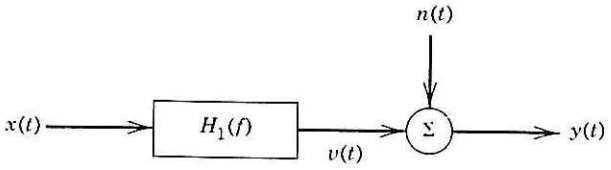
$$G_{vv}(f) = |H(f)|^2 G_{uu}(f) \tag{6.35}$$

$$G_{uv}(f) = H(f)G_{uu}(f) \tag{6.36}$$

Various cases occur, depending on the correlation properties of $m(t)$ and $n(t)$ to each other and to the signals. Three cases of interest are

Case 1. No input noise; uncorrelated output noise
Case 2. No output noise; uncorrelated input noise
Case 3. Both noises present; uncorrelated with each other and with the signals

CASE 1



CASE 1. *No Input Noise; Uncorrelated Output Noise*

$$x(t) = u(t)$$

$$y(t) = v(t) + n(t) \qquad G_{vn}(f) = 0$$

$$G_{xx}(f) = G_{uu}(f)$$

$$G_{yy}(f) = G_{vv}(f) + G_{nn}(f)$$

$$G_{xy}(f) = G_{uv}(f) = H_1(f)G_{xx}(f)$$

$$H_1(f) = \frac{G_{xy}(f)}{G_{xx}(f)} \tag{6.37}$$

$$G_{vv}(f) = |H_1(f)|^2 G_{uu}(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)} \tag{6.38}$$

Note that $G_{vv}(f)$ can be calculated from $x(t)$ and $y(t)$ even though $v(t)$ cannot be measured. Also, $G_{nn}(f)$ can be calculated without measuring $n(t)$. For applications in practice, this is by far the most important case because one can often define inputs and minimize input noise. However, one will have no control over the output noise which is due to nonlinear operations or the contributions from other unmeasured inputs.

For Case 1, the ordinary coherence function is

$$\gamma_{xy}^2(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)G_{yy}(f)} = \frac{|G_{uv}(f)|^2}{G_{uu}(f)[G_{vv}(f) + G_{nn}(f)]}$$

$$= \frac{1}{1 + G_{nn}(f)/G_{vv}(f)} \tag{6.39}$$

since

$$\gamma_{uv}^2(f) = \frac{|G_{uv}(f)|^2}{G_{uu}(f)G_{vv}(f)} = 1$$

Note that $\gamma_{xy}^2(f) < 1$ when $G_{nn}(f) > 0$ with

$$G_{vv}(f) = \gamma_{xy}^2(f)G_{yy}(f) \tag{6.40}$$

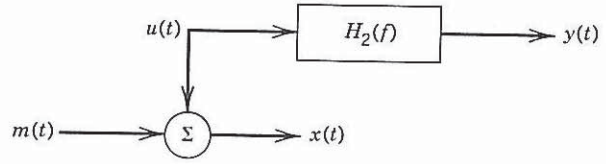
This product of $\gamma_{xy}^2(f)$ with $G_{yy}(f)$ is called the *coherent output spectrum*.

Note also that the *noise output spectrum* is

$$G_{nn}(f) = [1 - \gamma_{xy}^2(f)]G_{yy}(f) \tag{6.41}$$

Thus, $\gamma_{xy}^2(f)$ can be interpreted here as the portion of $G_{yy}(f)$ that is due to $x(t)$ at frequency f, while $[1 - \gamma_{xy}^2(f)]$ is a measure of the portion of $G_{yy}(f)$ not due to $x(t)$ at frequency f. Here, the ordinary coherence function decomposes the measured *output* spectrum into its uncorrelated components due to the input signal and extraneous noise. Equation (6.40) is the basis for solving many source identification problems using ordinary coherence functions, as illustrated in Reference 6.1.

CASE 2. *No Output Noise; Uncorrelated Input Noise*



$$x(t) = u(t) + m(t) \qquad G_{um}(f) = 0$$

$$y(t) = v(t)$$

$$G_{xx}(f) = G_{uu}(f) + G_{mm}(f)$$

$$G_{yy}(f) = G_{vv}(f) = |H_2(f)|^2 G_{uu}(f) = H_2(f)[H_2^*(f)G_{uu}(f)]$$

$$G_{xy}(f) = G_{uv}(f) = H_2(f)G_{uu}(f)$$

$$G_{yx}(f) = H_2^*(f)G_{uu}(f)$$

$$H_2(f) = \frac{G_{yy}(f)}{G_{yx}(f)} \tag{6.42}$$

$$G_{uu}(f) = \frac{G_{vv}(f)}{|H_2(f)|^2} = \frac{|G_{yx}(f)|^2}{G_{yy}(f)} = \frac{|G_{xy}(f)|^2}{G_{yy}(f)} \tag{6.43}$$

Here $G_{uu}(f)$ and $G_{mm}(f)$ can be determined from $x(t)$ and $y(t)$. This case is useful for special applications as discussed in Reference 6.2. It should not be applied when output noise is expected because of possible nonlinear operations, the contributions from other inputs, and/or output measurement instrumentation noise. Case 1 is always preferred in these situations.

For Case 2, the ordinary coherence function is

$$\gamma_{xy}^2(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)G_{yy}(f)} = \frac{|G_{uv}(f)|^2}{[G_{uu}(f) + G_{mm}(f)]G_{vv}(f)}$$

$$= \frac{1}{1 + [G_{mm}(f)/G_{uu}(f)]} \tag{6.44}$$

It follows that $\gamma_{xy}^2(f) < 1$ when $G_{mm}(f) > 0$ with

$$G_{uu}(f) = \gamma_{xy}^2(f)G_{xx}(f) \tag{6.45}$$

$$G_{mm}(f) = [1 - \gamma_{xy}^2(f)]G_{xx}(f) \tag{6.46}$$

Thus, for Case 2, the ordinary coherence function can decompose a measured *input* spectrum into its uncorrelated signal and noise components.

If one divides the frequency response function $H_1(f)$ for Case 1 as given by Equation (6.37) by the frequency response function $H_2(f)$ for Case 2 as given by Equation (6.42), the result is the coherence function, namely,

$$\frac{H_1(f)}{H_2(f)} = \frac{G_{xy}(f)/G_{xx}(f)}{G_{yy}(f)/G_{yx}(f)} = \gamma_{xy}^2(f) \tag{6.47}$$

CASE 3. ***Both Noises Present; Uncorrelated with Each Other and with the Signals***

$$x(t) = u(t) + m(t) \qquad G_{um}(f) = G_{vn}(f) = 0$$

$$y(t) = v(t) + n(t) \qquad G_{mn}(f) = 0$$

$$G_{xx}(f) = G_{uu}(f) + G_{mm}(f)$$

$$G_{yy}(f) = G_{vv}(f) + G_{nn}(f)$$

$$G_{xy}(f) = G_{uv}(f) = H(f)G_{uu}(f)$$

$$G_{vv}(f) = |H(f)|^2 G_{uu}(f) \tag{6.48}$$

Here, $H(f)$ cannot be determined from the measured $x(t)$ and $y(t)$ without a knowledge or measurement of the input noise. Specifically,

$$H(f) = \frac{G_{xy}(f)}{G_{uu}(f)} = \frac{G_{xy}(f)}{G_{xx}(f) - G_{mm}(f)} \tag{6.49}$$

$$|H(f)|^2 = \frac{G_{vv}(f)}{G_{uu}(f)} = \frac{G_{yy}(f) - G_{nn}(f)}{G_{xx}(f) - G_{mm}(f)} \tag{6.50}$$

Note that $H(f)$ is a function of $G_{mm}(f)$, but is independent of $G_{nn}(f)$ by Equation (6.49). Using Equation (6.50) to calculate $|H(f)|^2$ shows that $|H(f)|^2$ is a function of both $G_{mm}(f)$ and $G_{nn}(f)$.

For Case 3, the ordinary coherence function is

$$\gamma_{xy}^2(f) = \frac{|G_{xy}(f)|^2}{G_{xx}(f)G_{yy}(f)} = \frac{|G_{uv}(f)|^2}{[G_{uu}(f) + G_{mm}(f)][G_{vv}(f) + G_{nn}(f)]}$$

$$= \frac{1}{1 + \alpha(f) + \beta(f) + \alpha(f)\beta(f)} \tag{6.51}$$

where $\alpha(f)$ and $\beta(f)$ are the noise-to-signal ratios given by

$$\alpha(f) = [G_{mm}(f)/G_{uu}(f)]$$

$$\beta(f) = [G_{nn}(f)/G_{vv}(f)] \tag{6.52}$$

Clearly, $\gamma_{xy}^2(f) < 1$ whenever $\alpha(f) > 0$ or $\beta(f) > 0$. Here, using only $x(t)$ and $y(t)$, it is *not* possible to decompose $G_{xx}(f)$ or $G_{yy}(f)$ into their separate signal and noise components.

6.1.4 *Optimum Frequency Response Functions*

Return now to the basic single-input/single-output system with output noise only (Case 1), as illustrated in Figure 6.7. Without assuming that $n(t)$ is uncorrelated with $v(t)$, let $H(f)$ be *any* linear frequency response function acting on $x(t)$. Of interest is the specific $H(f)$ that will minimize the noise at the output; that is, the optimum estimate of $H(f)$ in the least squares sense. Note that if there is any correlation between the output signal and noise, as would occur if the output included signal-dependent instrumentation noise or the contributions of other inputs that are correlated with the measured $x(t)$, then the resulting optimum $H(f)$ will not represent the physical direct path between the points where $x(t)$ and $y(t)$ are measured. Also, there may be nonlinear operations between the measured input and output data. In any case, the optimum $H(f)$ will simply constitute a mathematical function that defines the best linear relationship between $x(t)$ and $y(t)$ in the least squares sense.

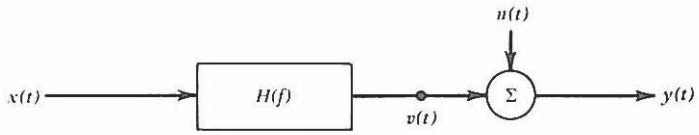


Figure 6.7 Single-input/single-output system with output noise.

For any set of long records of finite length T in Figure 6.7, the governing relation is

$$Y(f) = H(f)X(f) + N(f) \tag{6.53}$$

where capital letters are finite Fourier transforms of associated time domain records. Solving for $N(f)$ and $N^*(f)$ gives

$$N(f) = Y(f) - H(f)X(f) \qquad N^*(f) = Y^*(f) - H^*(f)X^*(f)$$

It follows that

$$|N(f)|^2 = |Y(f)|^2 - H(f)Y^*(f)X(f)$$
$$-H^*(f)X^*(f)Y(f) + H(f)H^*(f)|X(f)|^2 \tag{6.54}$$

Taking the expectation of Equation (6.54), multiplying by $(2/T)$, and letting T increase to infinity yields

$$G_{nn}(f) = G_{yy}(f) - H(f)G_{yx}(f) - H^*(f)G_{xy}(f) + H(f)H^*(f)G_{xx}(f) \tag{6.55}$$

This is the form of $G_{nn}(f)$ for *any* $H(f)$. By definition, the *optimum* $H(f)$ will now be defined as that $H(f)$ that minimizes $G_{nn}(f)$ over all possible choices of $H(f)$. This is called the *least-squares estimate*, defined in Section 4.8.2.

The minimization of $G_{nn}(f)$ as a function of $H(f)$ will now be carried out. To simplify the derivation, the dependence on f will be omitted. Thus

$$G_{nn} = G_{yy} - HG_{yx} - H^*G_{xy} + HH^*G_{xx} \tag{6.56}$$

Now let the complex numbers be expressed in terms of their real and imaginary parts as follows:

$$H = H_R - jH_I \qquad H^* = H_R + jH_I$$

$$G_{xy} = G_R - jG_I \qquad G_{yx} = G_R + jG_I$$

Then

$$G_{nn} = G_{yy} - (H_R - jH_I)G_{yx} - (H_R + jH_I)G_{xy} + (H_R^2 + H_I^2)G_{xx}$$

To find the form of H that will minimize G_{nn}, one should now set the partial derivatives of G_{nn} with respect to H_R and H_I equal to zero and solve the resulting pair of equations. This gives

$$\frac{\partial G_{nn}}{\partial H_R} = -G_{yx} - G_{xy} + 2H_R G_{xx} = 0$$

$$\frac{\partial G_{nn}}{\partial H_I} = jG_{yx} - jG_{xy} + 2H_I G_{xx} = 0$$

which leads to

$$H_R = \frac{G_{xy} + G_{yx}}{2G_{xx}} = \frac{G_R}{G_{xx}}$$

$$H_I = \frac{j(G_{xy} - G_{yx})}{2G_{xx}} = \frac{G_I}{G_{xx}}$$

Hence the optimum H is

$$H = H_R - jH_I = \frac{G_R - jG_I}{G_{xx}} = \frac{G_{xy}}{G_{xx}} \tag{6.57}$$

Again, the optimum H calculated by Equation (6.57), using arbitrary measured records, does not have to be physically realizable; it may be only a theoretically computed result.

Another important property satisfied by the optimum $H(f)$ is revealed by substitution of the optimum system satisfying Equation (6.57) into Equation (6.55). This gives the noise output spectrum

$$G_{nn}(f) = \left[1 - \gamma_{xy}^2(f)\right]G_{yy}(f) \tag{6.58}$$

which leads to the coherent output spectrum

$$G_{vv}(f) = G_{yy}(f) - G_{nn}(f) = \gamma_{xy}^2(f)G_{yy}(f) \tag{6.59}$$

Moreover, using the optimum $H(f)$ shows that

$$G_{xv}(f) = H(f)G_{xx}(f)$$

and

$$G_{xn}(f) = G_{xy}(f) - H(f)G_{xx}(f) = 0 \tag{6.60}$$

It follows that

$$G_{vn}(f) = H^*(f)G_{xn}(f) = 0 \tag{6.61}$$

Thus $n(t)$ and $v(t)$ will *automatically be uncorrelated* when the optimum $H(f)$ is used to estimate the linear system in Figure 6.7.

REFERENCES

6.1 Bendat, J. S., and Piersol, A. G., *Engineering Applications of Correlation and Spectral Analysis*, Wiley-Interscience, New York, 1980.

6.2 Upton, R., "Innovative Functions for Two-Channel FFT Analyzers," *Sound and Vibration*, Vol. 18, p. 18, March 1984.

6.3 Chung, J. Y., "Rejection of Flow Noise Using a Coherence Function Method," *Journal of the Acoustical Society of America*, Vol. 62, p. 388, 1977.

6.4 Adams, E. P., *Smithsonian Mathematical Formulae and Tables of Elliptic Functions*, Smithsonian Institution, Washington, D.C., 1947.

CHAPTER 7

MULTIPLE-INPUT / OUTPUT RELATIONSHIPS

Material contained in Chapter 6 is now extended to multiple-input problems. As before, records are assumed to be from stationary random processes with zero mean values, and systems are constant-parameter linear systems. The discussion includes multiple-input/single-output models and multiple-input/multiple-output models. Partial and multiple coherence functions are defined for these models. Iterative computational procedures are explained that provide physical insight to decompose the models, and are also more efficient than other matrix techniques. The various models considered in Chapters 6 and 7 represent useful building blocks to analyze more complicated physical systems that can be broken down into such combinations of basic subsystems.

7.1 MULTIPLE-INPUT/SINGLE-OUTPUT MODELS

This section describes multiple-input/single-output analysis problems as previously discussed and solved in References 7.1–7.3. Later material gives recommended procedures and interpretation of the various results. Consider q constant-parameter linear systems $H_i(f)$, $i = 1, 2, \ldots, q$, with q clearly defined measurable inputs $x_i(t)$, $i = 1, 2, \ldots, q$, and one measured output $y(t)$, as illustrated in Figure 7.1. There is *no* requirement that the inputs be mutually uncorrelated. The output noise term $n(t)$ accounts for all deviations from the ideal model, which may be due to unmeasured inputs, nonlinear operations, nonstationary effects, and instrument noise. Measured records are assumed to be realizations of stationary (ergodic) random processes, where nonzero mean values and any periodicities have been removed prior to this analysis. Situations of multiple-input/multiple-output models are direct extensions of these techniques by merely considering such models to be combinations of the multiple-input/single-output cases of Figure 7.1.

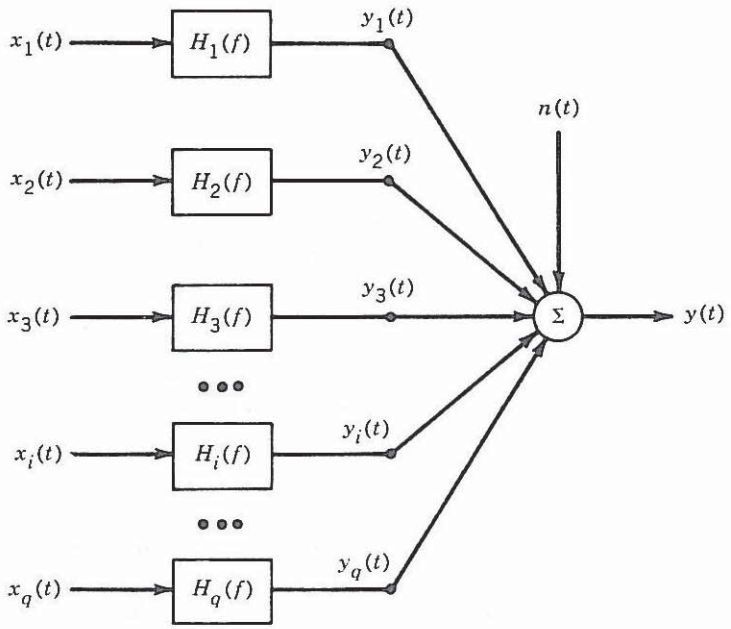


Figure 7.1 Multiple-input/single-output system.

7.1.1 *General Relationships*

Referring to the multiple-input model in Figure 7.1, four conditions are required for this model to be well defined.

1. None of the ordinary coherence functions between any pair of input records should equal unity. If this occurs, these two inputs contain redundant information and one of the inputs should be eliminated from the model. This consideration allows distributed input systems to be studied as discrete inputs.
2. None of the ordinary coherence functions between any input and the total output should equal unity. If this occurs, then the other inputs are not contributing to this output and the model should be considered as simply a single-input/single-output model.
3. The multiple coherence function (defined later) between any input and other inputs, excluding the given input, should not equal unity. If this occurs, then this input can be obtained by linear operations from the other inputs. This input is not providing any new information to the output and should be eliminated from the model.
4. The multiple coherence function between the output and the given inputs, in a practical situation, should be sufficiently high, say above 0.50, for the theoretical assumptions and later conclusions to be reasonable. Otherwise, some important inputs are probably being omitted, or nonlinear effects should be considered. This value of 0.50 is not precise, but is a matter of engineering and statistical judgment, based on the physical environment and the amount of data available for analysis.

It is assumed that one can make simultaneous measurements of the input and output time-history records in any specified multiple-input/output model. It is assumed also that possible system errors and statistical errors in computed quantities have been minimized by careful calibration and choice of data processing parameters. In particular, it is required that one obtain good estimates of the real-valued autospectral density functions of each record and the complex-valued cross-spectral density functions between every pair of records. From these stored spectral quantities, the following results are desired:

1. Decomposition of the output spectra into physically meaningful components due to the measured input records
2. Determination of optimum constant-parameter linear systems between each input and the output to minimize any noise spectra at the output that is not due to linear operations from the measured input records

The output $y(t)$ may be considered to be the sum of the unmeasured q outputs $y_i(t)$, $i = 1, 2, \ldots, q$, plus the noise term $n(t)$, namely,

$$y(t) = \sum_{i=1}^{q} y_i(t) + n(t) \tag{7.1}$$

with corresponding finite Fourier transforms

$$Y(f) = \sum_{i=1}^{q} Y_i(f) + N(f) \tag{7.2}$$

Each output term $Y_i(f)$ for $i = 1, 2, \ldots, q$ satisfies

$$Y_i(f) = H_i(f) X_i(f) \tag{7.3}$$

Thus, the basic frequency domain equation for Figure 7.1 is

$$Y(f) = \sum_{i=1}^{q} H_i(f) X_i(f) + N(f) \tag{7.4}$$

where each $X_i(f)$ and $Y(f)$ can be computed from the measured $x_i(t)$ and $y(t)$. From this information it is required to determine the systems $H_i(f)$ and other quantities of interest where all results are functions of frequency.

Finite Fourier transforms $X_i(f)$ and $Y(f)$ for single records $x_i(t)$ and $y(t)$ of length T are

$$X_i(f) = \int_0^T x_i(t)e^{-j2\pi ft}\,dt \qquad Y(f) = \int_0^T y(t)e^{-j2\pi ft}\,dt \tag{7.5}$$

As per Equations (5.66) and (5.67), one-sided autospectral and cross-spectral density functions are defined by

$$G_{ii}(f) = G_{x_ix_i}(f) = \lim_{T\to\infty} \frac{2}{T} E\left[|X_i(f)|^2\right]$$

$$G_{ij}(f) = G_{x_ix_j}(f) = \lim_{T\to\infty} \frac{2}{T} E\left[X_i^*(f)X_j(f)\right] \tag{7.6}$$

$$G_{yy}(f) = \lim_{T\to\infty} \frac{2}{T} E\left[|Y(f)|^2\right]$$

$$G_{iy}(f) = G_{x_iy}(f) = \lim_{T\to\infty} \frac{2}{T} E\left[X_i^*(f)Y(f)\right] \tag{7.7}$$

These one-sided functions where $G(f) = 0$ for $f < 0$ can be replaced by theoretical two-sided functions $S(f)$, as discussed in Section 5.2.1. All formulas in this chapter can be stated in terms of either $G(f)$ or $S(f)$ quantities on both sides of the equations. The ratio of two $G(f)$ quantities is the same as the ratio of the corresponding two $S(f)$ quantities. Further work here, as in Chapter 6, will use the one-sided $G(f)$ quantities.

In practice, one obtains only estimates of Equations (7.6) and (7.7) since T will be finite and the expected value operation $E[\ ]$ can be taken only over a finite number of sample records. Also, when records are digitized, as discussed in Chapters 10 and 11, results will be obtained only at selected discrete frequencies. At any such frequency f, an estimate of $G_{xy}(f)$, denoted $\hat{G}_{xy}(f)$, is usually obtained by

$$\hat{G}_{xy}(f) = \frac{2}{n_d T} \sum_{k=1}^{n_d} X_k^*(f)Y_k(f) \tag{7.8}$$

where n_d is the number of different (disjoint) sample records of $x(t)$ and $y(t)$, each of length T, so that the total record length $T_r = n_dT$. It is proved in Chapter 8 that to reduce bias errors, T should be made as large as possible, while to reduce random errors, n_d should be as large as possible. Hence compromise choices are necessary when T_r is fixed.

If desired, corresponding time domain equations can be written involving convolution integrals of the respective weighting functions $h_i(\tau)$, $i = 1, 2, \ldots, q$, associated with the $H_i(f)$. In place of Equation (7.3), one would have

$$y_i(t) = \int_0^\infty h_i(\tau)x_i(t-\tau)\,d\tau \tag{7.9}$$

where the lower limit is zero only when the systems are physically realizable. Such convolution integrals and their extensions to correlation functions are much more complicated than associated Fourier transform spectral relations and will not be used in the further development here. These correlation relations do not show how results vary as a function of frequency. That such information is hidden within the correlation functions is no justification for computing them when it is unnecessary and more efficient to compute spectral quantities directly.

7.1.2 *General Case of Arbitrary Inputs*

Consider the general case of arbitrary inputs. From Equation (7.4), with a different index of summation j instead of i,

$$Y(f) = \sum_{j=1}^{q} H_j(f)X_j(f) + N(f) \tag{7.10}$$

Multiplication of both sides by $X_i^*(f)$ for any fixed $i = 1, 2, \ldots, q$ yields

$$X_i^*(f)Y(f) = \sum_{j=1}^{q} H_j(f)X_i^*(f)X_j(f) + X_i^*(f)N(f)$$

Expected values of both sides shows that

$$E\left[X_i^*(f)Y(f)\right] = \sum_{j=1}^{q} H_j(f)E\left[X_i^*(f)X_j(f)\right] + E\left[X_i^*(f)N(f)\right]$$

Equations (7.6) and (7.7), with a scale factor of $(2/T)$ to obtain one-sided results, now proves at any f that

$$G_{iy}(f) = \sum_{j=1}^{q} H_j(f)G_{ij}(f) + G_{in}(f) \qquad i = 1, 2, \ldots, q$$

where the cross-spectral terms $G_{in}(f)$ will be zero if $n(t)$ is uncorrelated with each $x_i(t)$. Making this assumption gives the set of equations

$$G_{iy}(f) = \sum_{j=1}^{q} H_j(f)G_{ij}(f) \qquad i = 1, 2, \ldots, q \tag{7.11}$$

This is a set of q equations in q unknowns, the $H_i(f)$ for $i = 1, 2, \ldots, q$, where

all the spectral terms shown are computed from the measured input and output records. If the model is well defined, one can solve for the $H_i(f)$ by matrix techniques using Cramer's rule or the equivalent.

In terms of the computed $H_i(f)$, as found by solving Equation (7.11), the total output autospectral density function $G_{yy}(f)$ is

$$G_{yy}(f) = \sum_{i=1}^{q} \sum_{j=1}^{q} H_i^*(f) H_j(f) G_{ij}(f) + G_{nn}(f) \tag{7.12}$$

assuming, as before, that $n(t)$ is uncorrelated with each $x_i(t)$. Equation (7.12) is derived from

$$Y^*(f) = \sum_{i=1}^{q} H_i^*(f) X_i^*(f) + N^*(f)$$

$$Y^*(f)Y(f) = \left[\sum_{i=1}^{q} H_i^*(f) X_i^*(f) + N^*(f)\right]\left[\sum_{j=1}^{q} H_j(f) X_j(f) + N(f)\right]$$

$$= \sum_{i=1}^{q} \sum_{j=1}^{q} H_i^*(f) H_j(f) X_i^*(f) X_j(f) + N^*(f) N(f)$$

$$+ \sum_{i=1}^{q} H_i^*(f) X_i^*(f) N(f) + \sum_{j=1}^{q} H_j(f) N^*(f) X_j(f)$$

Expected values of both sides, multiplication of both sides by the scale factor $(2/T)$, and passage to the limit gives the result

$$G_{yy}(f) = \sum_{i=1}^{q} \sum_{j=1}^{q} H_i^*(f) H_j(f) G_{ij}(f) + G_{nn}(f)$$

$$+ \sum_{i=1}^{q} H_i^*(f) G_{in}(f) + \sum_{j=1}^{q} H_j(f) G_{nj}(f)$$

This reduces to Equation (7.12) when the cross-spectral terms $G_{in}(f) = 0$ for all f and all i. For q different inputs and for terms $G_{ij}(f) \neq 0$ when $i \neq j$, the output $G_{yy}(f)$ of Equation (7.12) contains $(q^2 + 1)$ parts. Decomposing $G_{yy}(f)$ into each of these parts and interpreting their meaning can be a very involved exercise.

7.1.3 *Special Case of Mutually Uncorrelated Inputs*

Consider the important special case when not only is $n(t)$ uncorrelated with each of the $x_i(t)$, but the inputs are mutually uncorrelated with each other. For this special case, Equations (7.11) and (7.12) become

$$G_{iy}(f) = H_i(f) G_{ii}(f) \qquad i = 1, 2, \ldots, q \tag{7.13}$$

$$G_{yy}(f) = \sum_{i=1}^{q} |H_i(f)|^2 G_{ii}(f) + G_{nn}(f) \tag{7.14}$$

These relations have a very simple interpretation as a collection of distinct single-input/single-output models. No system of equations is required to solve for the $H_i(f)$. From Equation (7.13), each $H_i(f)$ is given by the ratio

$$H_i(f) = \frac{G_{iy}(f)}{G_{ii}(f)} \qquad i = 1, 2, \ldots, q \tag{7.15}$$

The output $G_{yy}(f)$ of Equation (7.14), unlike the output of Equation (7.12), now contains only $(q + 1)$ parts. Each of the q parts is the coherent output spectrum result of single-input/single-output models, namely,

$$|H_i(f)|^2 G_{ii}(f) = \gamma_{iy}^2(f) G_{yy}(f) \tag{7.16}$$

These quantities represent the input record $x_i(t)$ passing only through its particular $H_i(f)$ to reach $y(t)$. No leakage of $x_i(t)$ occurs through any other system $H_j(f)$ since $x_i(t)$ is uncorrelated with $x_j(t)$ for $i \neq j$.

For general cases of multiple input records, with arbitrary correlations between the inputs, any input record $x_i(t)$ can reach $y(t)$ by passage through any of the systems $H_i(f)$ for all $i = 1, 2, \ldots, q$. It can now be quite difficult to decompose $G_{yy}(f)$ into its respective contributions from each of the input records if one attempts to solve the general multiple-input/output problem by the usual "brute force" matrix techniques. These matrix techniques also provide no physical insight into how models should be formulated to provide good agreements between mathematical results and physical situations. Better methods are required to solve many problems, as explained in this chapter.

7.2 TWO-INPUT/ONE-OUTPUT MODELS

For an understanding of the methodology to solve general multiple-input/output problems, one should consider special cases of two-input/one-output models and three-input/one-output models. The general case is then merely more of the same features encountered in these special cases. This will now be done in detail for the two-input/one-output model. Equations for the three-input/one-output model can be developed from the material in Section 7.3.

7.2.1 *Basic Relationships*

Consider a two-input/one-output model as pictured in Figure 7.2, where the two inputs may be correlated. Assume that $x_1(t)$, $x_2(t)$, and $y(t)$ can be

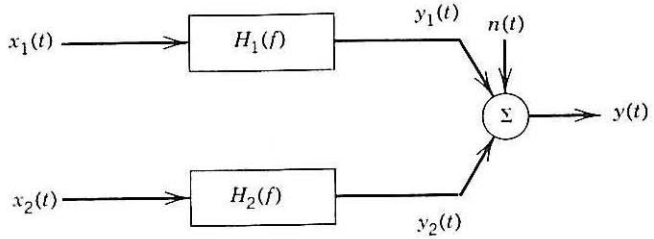


Figure 7.2 Two-input/one-output model.

measured. This system is defined by the basic transform relation

$$Y(f) = H_1(f)X_1(f) + H_2(f)X_2(f) + N(f) \tag{7.17}$$

For arbitrary $n(t)$, which may be correlated with $x_1(t)$ and/or $x_2(t)$, the one-sided autospectral density function of $y(t)$ from Equations (7.5)–(7.8), for a very long but finite record length T, is given by

$$G_{yy}(f) = G_{yy} = \frac{2}{T}E[Y^*Y]$$

$$= \frac{2}{T}E[\{H_1^*X_1^* + H_2^*X_2^* + N^*\}\{H_1X_1 + H_2X_2 + N\}] \tag{7.18}$$

where the dependence on f and the limiting operation on T have been omitted to simplify the notation. Equation (7.18) becomes

$$G_{yy} = |H_1|^2G_{11} + H_1^*H_2G_{12} + H_1H_2^*G_{21} + |H_2|^2G_{22} + G_{nn}$$
$$+ H_1^*G_{1n} + H_1G_{n1} + H_2^*G_{2n} + H_2G_{n2} \tag{7.19}$$

where the last four terms occur only when $n(t)$ is correlated with $x_1(t)$ and $x_2(t)$.

For arbitrary $n(t)$, the cross-spectral density functions between $x_1(t)$ and $y(t)$, and between $x_2(t)$ and $y(t)$, are given in a similar way by

$$G_{1y} = \frac{2}{T}E[X_1^*Y] = \frac{2}{T}E[X_1^*(H_1X_1 + H_2X_2 + N)]$$
$$= H_1G_{11} + H_2G_{12} + G_{1n} \tag{7.20}$$

$$G_{2y} = \frac{2}{T}E[X_2^*Y] = \frac{2}{T}E[X_2^*(H_1X_1 + H_2X_2 + N)]$$
$$= H_1G_{21} + H_2G_{22} + G_{2n} \tag{7.21}$$

where, again, the dependence on f and the limiting operation on T have been omitted to simplify the notation. The terms G_{1n} and G_{2n} occur only when $n(t)$ is correlated with $x_1(t)$ and $x_2(t)$.

From knowledge of $x_1(t)$, $x_2(t)$, and $y(t)$, the one-sided spectral quantities will be

$$G_{11} = G_{11}(f) = \text{autospectrum of } x_1(t)$$

$$G_{22} = G_{22}(f) = \text{autospectrum of } x_2(t)$$

$$G_{yy} = G_{yy}(f) = \text{autospectrum of } y(t)$$

$$G_{12} = G_{12}(f) = \text{cross-spectrum between } x_1(t) \text{ and } x_2(t)$$

$$G_{1y} = G_{1y}(f) = \text{cross-spectrum between } x_1(t) \text{ and } y(t)$$

$$G_{2y} = G_{2y}(f) = \text{cross-spectrum between } x_2(t) \text{ and } y(t)$$

$$G_{21} = G_{12}^*(f) = \text{complex conjugate of } G_{12}$$

All spectral quantities involving $n(t)$ will be unknown. In particular, Equations (7.20) and (7.21) cannot be solved for H_1 and H_2 unless the terms $G_{1n} = 0$ and $G_{2n} = 0$, where

$$G_{1n} = G_{1n}(f) = \text{cross-spectrum between } x_1(t) \text{ and } n(t)$$

$$G_{2n} = G_{2n}(f) = \text{cross-spectrum between } x_2(t) \text{ and } n(t)$$

When $n(t)$ is uncorrelated with $x_1(t)$ and $x_2(t)$, Equations (7.20) and (7.21) become

$$G_{1y}(f) = H_1(f)G_{11}(f) + H_2(f)G_{12}(f)$$
$$G_{2y}(f) = H_1(f)G_{21}(f) + H_2(f)G_{22}(f) \tag{7.22}$$

The solutions for $H_1(f)$ and $H_2(f)$, assuming $\gamma_{12}^2(f) \neq 1$, are

$$H_1(f) = \frac{G_{1y}(f)\left[1 - \dfrac{G_{12}(f)G_{2y}(f)}{G_{22}(f)G_{1y}(f)}\right]}{G_{11}(f)[1 - \gamma_{12}^2(f)]}$$

$$H_2(f) = \frac{G_{2y}(f)\left[1 - \dfrac{G_{21}(f)G_{1y}(f)}{G_{11}(f)G_{2y}(f)}\right]}{G_{22}(f)[1 - \gamma_{12}^2(f)]} \tag{7.23}$$

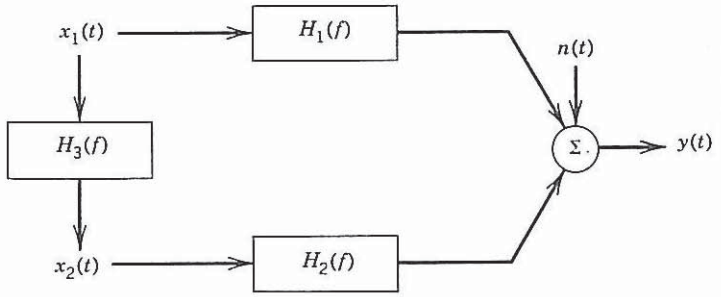


Figure 7.3 Illustration of fully coherent inputs.

where the ordinary coherence function

$$\gamma_{12}^2(f) = \frac{|G_{12}(f)|^2}{G_{11}(f)G_{22}(f)} \tag{7.24}$$

For the special case of uncorrelated inputs when $\gamma_{12}^2(f) = 0$, the terms $G_{12}(f)$ and $G_{21}(f)$ are zero also, and Equation (7.23) reduces to the usual relations for single-input/single-output models

$$H_1(f) = \frac{G_{1y}(f)}{G_{11}(f)} \qquad H_2(f) = \frac{G_{2y}(f)}{G_{22}(f)} \tag{7.25}$$

The case in which $\gamma_{12}^2(f) = 1$ must be handled separately. A coherence function of unity between $x_1(t)$ and $x_2(t)$ implies complete linear dependence. Hence one would consider a linear system existing between them, as illustrated in Figure 7.3. The implication is that the first input $x_1(t)$ is actually taking two different paths to arrive at the output $y(t)$. For this situation, a single frequency response function $H(f)$ will relate $y(t)$ to $x_1(t)$, namely,

$$H(f) = H_1(f) + H_2(f)H_3(f)$$

When $n(t)$ is uncorrelated with $x_1(t)$ and $x_2(t)$, but $G_{12}(f) \neq 0$, Equation (7.19) becomes

$$G_{yy}(f) = |H_1(f)|^2 G_{11}(f) + H_1^*(f)H_2(f)G_{12}(f)$$
$$+ H_2^*(f)H_1(f)G_{21}(f) + |H_2(f)|^2 G_{22}(f) + G_{nn}(f) \tag{7.26}$$

This can be written as

$$G_{yy}(f) = G_{vv}(f) + G_{nn}(f) = G_{y:x}(f) + G_{y:n}(f) \tag{7.27}$$

where $G_{nn}(f) = G_{y:n}(f)$ is the output spectrum due to the noise, and $G_{vv}(f) = G_{y:x}(f)$ represents the ideal output spectrum due to the two inputs, as computed by the first four terms in Equation (7.26). To be explicit, $G_{vv}(f)$ comes from knowledge of the terms $G_{11}(f)$, $G_{12}(f)$, $G_{22}(f)$ plus the computed $H_1(f)$ and $H_2(f)$ by the general Equation (7.23). Finally, even though $n(t)$ is unmeasured, the autospectral density function $G_{nn}(f)$ can be computed by the formula

$$G_{nn}(f) = G_{yy}(f) - G_{vv}(f) \tag{7.28}$$

For the special case of uncorrelated inputs where $G_{12}(f) = 0$, the ideal output spectrum reduces to

$$G_{vv}(f) = |H_1(f)|^2 G_{11}(f) + |H_2(f)|^2 G_{22}(f) \tag{7.29}$$

where $H_1(f)$ and $H_2(f)$ are given by the special Equation (7.25). Here

$$|H_1(f)|^2 G_{11}(f) = \gamma_{1y}^2(f)G_{yy}(f)$$
$$|H_2(f)|^2 G_{22}(f) = \gamma_{2y}^2(f)G_{yy}(f) \tag{7.30}$$

where $\gamma_{1y}^2(f)$ and $\gamma_{2y}^2(f)$ are the ordinary coherence functions

$$\gamma_{1y}^2(f) = \frac{|G_{1y}(f)|^2}{G_{11}(f)G_{yy}(f)} \qquad \gamma_{2y}^2(f) = \frac{|G_{2y}(f)|^2}{G_{22}(f)G_{yy}(f)} \tag{7.31}$$

It follows for uncorrelated inputs that

$$G_{vv}(f) = \left[\gamma_{1y}^2(f) + \gamma_{2y}^2(f)\right] G_{yy}(f) \tag{7.32}$$

and

$$G_{nn}(f) = \left[1 - \gamma_{1y}^2(f) - \gamma_{2y}^2(f)\right] G_{yy}(f) \tag{7.33}$$

The quantity $|H_1|^2 G_{11}$ represents the spectral output of x_1 through the H_1 system only. Similarly, $|H_2|^2 G_{22}$ represents the spectral output of x_2 through the H_2 system only. These two results are the ordinary coherent output spectra $\gamma_{1y}^2 G_{yy}$ and $\gamma_{2y}^2 G_{yy}$, respectively, between input x_1 and the output y, and between input x_2 and the output y.

7.2.2 *Optimum Frequency Response Functions*

Equations (7.22) and (7.26) are based on the assumption that $n(t)$ is uncorrelated with $x_1(t)$ and $x_2(t)$. Without making this assumption, when $x_1(t)$ and

$x_2(t)$ pass through any pair of constant parameter linear systems $H_1(f)$ and $H_2(f)$, respectively, Equation (7.17) states that

$$N(f) = Y(f) - H_1(f)X_1(f) - H_2(f)X_2(f)$$

Hence, for any $H_1(f)$ and $H_2(f)$, the output noise spectrum is

$$\begin{aligned} G_{nn}(f) &= \frac{2}{T}E[N^*(f)N(f)] \\ &= G_{yy}(f) - H_1(f)G_{y1}(f) - H_2(f)G_{y2}(f) \\ &\quad - H_1^*(f)G_{1y}(f) + H_1^*(f)H_1(f)G_{11}(f) + H_1^*(f)H_2(f)G_{12}(f) \\ &\quad - H_2^*(f)G_{2y}(f) + H_1(f)H_2^*(f)G_{21}(f) + H_2(f)H_2^*(f)G_{22}(f) \end{aligned}$$

The *optimum frequency response functions* are now defined as the particular $H_1(f)$ and $H_2(f)$ that minimizes $G_{nn}(f)$ at any f over all possible choices of $H_1(f)$ and $H_2(f)$. They yield the optimum linear least squares prediction of $y(t)$ from $x_1(t)$ and $x_2(t)$.

To derive the optimum $H_1(f)$ and $H_2(f)$, as explained previously in Chapter 6, it is sufficient to set the following partial derivatives equal to zero.

$$\frac{\partial G_{nn}(f)}{\partial H_1^*(f)} = 0 \quad \text{holding } H_1(f) \text{ fixed}$$

$$\frac{\partial G_{nn}(f)}{\partial H_2^*(f)} = 0 \quad \text{holding } H_2(f) \text{ fixed}$$

This leads to the pair of equations

$$\begin{aligned} -G_{1y}(f) + H_1(f)G_{11}(f) + H_2(f)G_{12}(f) &= 0 \\ -G_{2y}(f) + H_1(f)G_{21}(f) + H_2(f)G_{22}(f) &= 0 \end{aligned}$$

which are identical to Equation (7.22).

For any pair, $H_1(f)$ and $H_2(f)$, the cross-spectral density functions $G_{1n}(f)$ and $G_{2n}(f)$ are given by

$$\begin{aligned} G_{1n}(f) &= \frac{2}{T}E[X_1^*(f)N(f)] \\ &= G_{1y}(f) - H_1(f)G_{11}(f) - H_2(f)G_{12}(f) \\ G_{2n}(f) &= \frac{2}{T}E[X_2^*(f)N(f)] \\ &= G_{2y}(f) - H_1(f)G_{21}(f) - H_2(f)G_{22}(f) \end{aligned}$$

When $H_1(f)$ and $H_2(f)$ are the optimum results that satisfy the relations of Equation (7.22), these two cross-spectra will be identically zero for all f. Thus, computation of the optimum $H_1(f)$ and $H_2(f)$ to minimize $G_{nn}(f)$ automatically makes $n(t)$ uncorrelated with $x_1(t)$ and $x_2(t)$, namely,

$$G_{1n}(f) = 0 \qquad G_{2n}(f) = 0$$

It should be pointed out here that the optimum computed $H_1(f)$ and $H_2(f)$ satisfying Equation (7.22) do not have to be physically realizable. That is, their associated weighting functions $h_1(\tau)$ and $h_2(\tau)$ may not be zero for τ less than zero. In fact, if the actual systems are nonlinear, these results will only represent optimum linear approximations as done by the computer and could never be the true nonlinear systems. Of course, if the actual systems are physically realizable constant-parameter linear systems where the model includes all the inputs producing the output, then these computed results would represent the true conditions provided the output noise is uncorrelated with all the inputs and there is negligible input noise. Henceforth, it will be assumed that $H_1(f)$ and $H_2(f)$ are computed by Equation (7.23). These results come from either (a) assuming in advance that $n(t)$ is uncorrelated with $x_1(t)$ and $x_2(t)$, or (b) requiring $H_1(f)$ and $H_2(f)$ to be the optimum constant-parameter linear systems to minimize $G_{nn}(f)$, whereupon the $n(t)$ in the model will become uncorrelated with $x_1(t)$ and $x_2(t)$.

7.2.3 *Ordinary and Multiple Coherence Functions*

Return to the general case of correlated inputs where $\gamma_{12}^2(f)$ is any positive value less than unity. The ordinary coherence functions between each input and the output are

$$\begin{aligned} \gamma_{1y}^2(f) &= \frac{|H_1(f)G_{11}(f) + H_2(f)G_{12}(f)|^2}{G_{11}(f)G_{yy}(f)} \\ \gamma_{2y}^2(f) &= \frac{|H_1(f)G_{21}(f) + H_2(f)G_{22}(f)|^2}{G_{22}(f)G_{yy}(f)} \end{aligned} \tag{7.34}$$

where the numerators represent $G_{1y}(f)$ and $G_{2y}(f)$ as computed using Equation (7.22). The product $\gamma_{1y}^2 G_{yy}$ still represents the ordinary coherent output spectrum between the input x_1 and the output y. However, x_1 does not now get to the output via the H_1 system only. Instead since $\gamma_{12}^2 \neq 0$, part of x_1 also gets to the output y via the H_2 system. Similarly, x_2 gets to the output via both H_1 and H_2 when $\gamma_{12}^2 \neq 0$, and the ordinary coherent output spectrum $\gamma_{2y}^2 G_{yy}$ represents all the ways x_2 can get to the output y. In general, for small $G_{nn}(f)$, the sum of $\gamma_{1y}^2(f)$ with $\gamma_{2y}^2(f)$ will be greater than unity when the inputs are correlated.

The *multiple coherence function* is a relatively simple concept and is a direct extension of the ordinary coherence function. By definition, the multiple coherence function is the ratio of the ideal output spectrum due to the measured inputs in the absence of noise to the total output spectrum, which includes the noise. In equation form, the multiple coherence function

$$\gamma^2_{y:x}(f) = \frac{G_{vv}(f)}{G_{yy}(f)} = 1 - \left[\frac{G_{nn}(f)}{G_{yy}(f)}\right] \tag{7.35}$$

since $G_{vv}(f) = G_{yy}(f) - G_{nn}(f)$. For the general two-input case, $G_{vv}(f)$ is shown in Equation (7.26). The *multiple coherent output spectrum* is defined by the product of the multiple coherence function and the output spectrum, namely,

$$G_{vv}(f) = \gamma^2_{y:x}(f)G_{yy}(f) \tag{7.36}$$

Clearly, for all values of f, Equation (7.35) shows

$$0 \le \gamma^2_{y:x}(f) \le 1 \tag{7.37}$$

The value of unity occurs when $G_{nn}(f) = 0$, indicating a perfect linear model, and the value of zero occurs when $G_{yy}(f) = G_{nn}(f)$, indicating that none of the output record comes from linear operations on the measured input records.

For a single-input/single-output model where $x_2(t) = 0$ and $H_2(f) = 0$, the ideal output spectrum as previously derived in Equation (6.59) is

$$G_{vv}(f) = |H(f)|^2 G_{xx}(f) = \gamma^2_{xy}(f)G_{yy}(f) \tag{7.38}$$

where $x_1(t) = x(t)$ and $H(f) = [G_{xy}(f)/G_{xx}(f)]$. It follows that

$$\gamma^2_{y:x}(f) = \frac{G_{vv}(f)}{G_{yy}(f)} = \gamma^2_{xy}(f) \tag{7.39}$$

In words, the multiple coherence function is the same here as the ordinary coherence function.

For a two-input/one-output model with uncorrelated inputs, the ideal output spectrum as previously derived in Equation (7.32) is

$$G_{vv}(f) = \left[\gamma^2_{1y}(f) + \gamma^2_{2y}(f)\right]G_{yy}(f)$$

Here, the multiple coherence function is

$$\gamma^2_{y:x}(f) = \gamma^2_{1y}(f) + \gamma^2_{2y}(f) \tag{7.40}$$

In words, for uncorrelated inputs, the multiple coherence function is the sum of the ordinary coherence functions between each input and the output. No such simple relation exists for correlated inputs.

Example 7.1. **Multiple Coherence for Two Uncorrelated Inputs.** Consider the example of an uncorrelated two-input/one-output system with negligible output noise, and assume that the two inputs produce equal output spectra. For this example,

$$\gamma^2_{y:x}(f) = \gamma^2_{1y}(f) + \gamma^2_{2y}(f) = 1.0$$

with

$$\gamma^2_{1y}(f) = \gamma^2_{2y}(f) = 0.50$$

Here there are ideal constant-parameter linear systems $H_1(f)$ and $H_2(f)$ between each input and the output. The fact that $\gamma^2_{1y}(f) = 0.50$ is due to the second input. If this was not known and one assumed a single-input/single-output model, erroneous conclusions would be drawn.

7.2.4 *Conditioned Spectral Density Functions*

When correlation exists between any pair of input records, to make mathematical results agree with the physical situation, wherever possible one should try to determine if one record causes part or all of the second record. If this can be determined, then turning off the first record will remove the correlated parts from the second record and leave only the part of the second record that is not due to the first record. To be precise, if engineering considerations state that any correlation between $x_1(t)$ and $x_2(t)$ comes from $x_1(t)$, then the optimum linear effects of $x_1(t)$ to $x_2(t)$ should be found, as denoted by $x_{2:1}(t)$. This should be subtracted from $x_2(t)$ to yield the *conditioned record* (also called the *residual record*) $x_{2\cdot1}(t)$ representing the part of $x_2(t)$ not due to $x_1(t)$. In equation form, $x_2(t)$ is decomposed into the sum of two uncorrelated terms shown in Figure 7.4, where

$$x_2(t) = x_{2:1}(t) + x_{2\cdot1}(t) \tag{7.41}$$

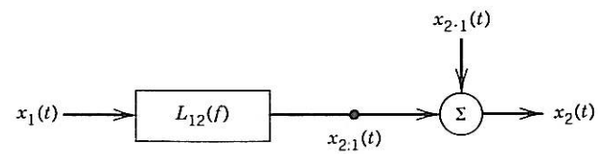


Figure 7.4 Decomposition of $x_2(t)$ from $x_1(t)$.

7.4 MATRIX FORMULAS FOR MULTIPLE-INPUT/MULTIPLE-OUTPUT MODELS

General cases will now be considered as represented in multiple-input/multiple-output models, where the number of output records is the same as the number of input records. It is straightforward to extend these results to cases of uneven numbers of input and output records. For greatest physical insight into these problems, it is recommended that they be broken down into multiple-input/single-output problems and solved by the algebraic procedures outlined in Sections 7.2 and 7.3. If one desires to solve these multiple-input/multiple-output models by employing matrix techniques, however, a consistent set of definitions and matrix formulas are stated in this section. These matrix results provide no useful decomposition of measured output spectra into their components from measured input records. Also, error analysis criteria for these matrix results are considerably more complicated than error analysis criteria for the results computed in Section 7.3, where a multiple-input/single-output model is changed into a set of ordered conditioned single-input/single-output models. Relatively simple error analysis criteria are developed in Chapter 9 for such single-input/single-output models.

7.4.1 *Multiple-Input/Multiple-Output Model*

$\mathbf{X}$ is a column vector representing the Fourier transforms of the q input records $X_i = X_i(f)$, $i = 1, 2, \ldots, q$, and $\mathbf{Y}$ is a column vector representing the Fourier transforms of the q output records $Y_k = Y_k(f)$, $k = 1, 2, \ldots, q$;

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_q \end{bmatrix} \qquad \mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_q \end{bmatrix} \tag{7.122}$$

$\mathbf{X}^*, \mathbf{Y}^*$ = complex conjugate (column) vectors of $\mathbf{X}, \mathbf{Y}$

$\mathbf{X}', \mathbf{Y}'$ = transpose (row) vectors of $\mathbf{X}, \mathbf{Y}$

$$\mathbf{G}_{xx} = \frac{2}{T} E\{\mathbf{X}^*\mathbf{X}'\} = \text{input spectral density matrix} \tag{7.123}$$

$$\mathbf{G}_{yy} = \frac{2}{T} E\{\mathbf{Y}^*\mathbf{Y}'\} = \text{output spectral density matrix} \tag{7.124}$$

$$\mathbf{G}_{xy} = \frac{2}{T} E\{\mathbf{X}^*\mathbf{Y}'\} = \text{input/output cross-spectral density matrix} \tag{7.125}$$

Ideally, Equations (7.123)–(7.125) involve a limit as $T \to \infty$, but the limit notation, like the frequency dependence notation, is omitted for clarity. In practice, with finite records, the limiting operation is never performed.

The basic matrix terms are defined as follows.

$$G_{ij} = \frac{2}{T} E\left[X_i^* X_j\right] \tag{7.126}$$

$$G_{y_i y_j} = \frac{2}{T} E\left[Y_i^* Y_j\right] \tag{7.127}$$

$$G_{iy_j} = \frac{2}{T} E\left[X_i^* Y_j\right] \tag{7.128}$$

$$\mathbf{G}_{xx} = \frac{2}{T} E\left\{ \begin{bmatrix} X_1^* \\ X_2^* \\ \vdots \\ X_q^* \end{bmatrix} \begin{bmatrix} X_1 & X_2 & \cdots & X_q \end{bmatrix} \right\}$$

$$= \begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1q} \\ G_{21} & G_{22} & \cdots & G_{2q} \\ \vdots & \vdots & & \vdots \\ G_{q1} & G_{q2} & \cdots & G_{qq} \end{bmatrix} \tag{7.129}$$

$$\mathbf{G}_{yy} = \frac{2}{T} E\left\{ \begin{bmatrix} Y_1^* \\ Y_2^* \\ \vdots \\ Y_q^* \end{bmatrix} \begin{bmatrix} Y_1 & Y_2 & \cdots & Y_q \end{bmatrix} \right\}$$

$$= \begin{bmatrix} G_{y_1 y_1} & G_{y_1 y_2} & \cdots & G_{y_1 y_q} \\ G_{y_2 y_1} & G_{y_2 y_2} & \cdots & G_{y_2 y_q} \\ \vdots & & & \\ G_{y_q y_1} & G_{y_q y_2} & \cdots & G_{y_q y_q} \end{bmatrix} \tag{7.130}$$

Observe that $\mathbf{G}_{xx}$ and $\mathbf{G}_{yy}$ are Hermitian matrices, namely, $G_{ij} = G_{ji}^*$ for all i

and j. For these Hermitian matrices, $\mathbf{G}_{xx}^{*} = \mathbf{G}_{xx}'$ and $\mathbf{G}_{yy}^{*} = \mathbf{G}_{yy}'$.

$$\mathbf{G}_{xy} = \frac{2}{T} E\left\{ \begin{bmatrix} X_1^* \\ X_2^* \\ \vdots \\ X_q^* \end{bmatrix} \begin{bmatrix} Y_1 & Y_2 & \cdots & Y_q \end{bmatrix} \right\}$$

$$= \begin{bmatrix} G_{1y_1} & G_{1y_2} & \cdots & G_{1y_q} \\ G_{2y_1} & G_{2y_2} & \cdots & G_{2y_q} \\ \vdots & \vdots & & \vdots \\ G_{qy_1} & G_{qy_2} & \cdots & G_{qy_q} \end{bmatrix} \tag{7.131}$$

In all these terms, $G_{iy_j} = G_{x_i y_j}(f)$, where the input precedes the output.

Define the system matrix between $\mathbf{X}$ and $\mathbf{Y}$ by $\mathbf{H}_{xy} = \mathbf{H}_{xy}(f)$, where, as above, *input always precedes output*. The matrix terms $H_{iy_k} = H_{x_i y_k}$. Then

$$\mathbf{H}_{xy} = \begin{bmatrix} H_{1y_1} & H_{1y_2} & \cdots & H_{1y_q} \\ H_{2y_1} & H_{2y_2} & \cdots & H_{2y_q} \\ \vdots & \vdots & & \vdots \\ H_{qy_1} & H_{qy_2} & \cdots & H_{qy_q} \end{bmatrix} \tag{7.132}$$

From this definition, it follows that

$$\mathbf{Y} = \mathbf{H}_{xy}'\mathbf{X} \tag{7.133}$$

where $\mathbf{H}_{xy}'$ is the *transpose* matrix to $\mathbf{H}_{xy}$. Thus

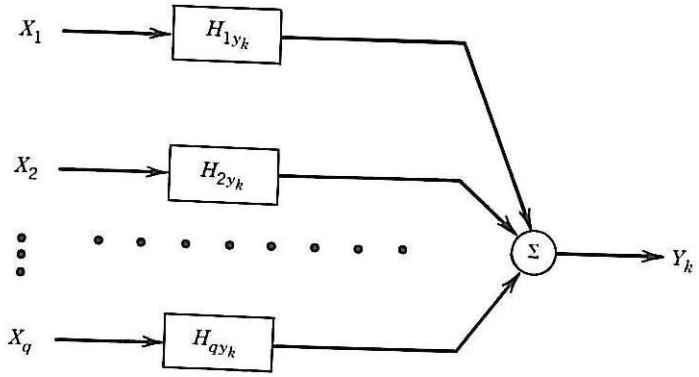
$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_q \end{bmatrix} = \begin{bmatrix} H_{1y_1} & H_{2y_1} & \cdots & H_{qy_1} \\ H_{1y_2} & H_{2y_2} & \cdots & H_{qy_2} \\ \vdots & \vdots & & \vdots \\ H_{1y_q} & H_{2y_q} & \cdots & H_{qy_q} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_q \end{bmatrix} \tag{7.134}$$

Note that this gives the algebraic results

$$Y_k = \sum_{i=1}^{q} H_{iy_k} X_i \qquad k = 1, 2, \ldots, q \tag{7.135}$$

This is the logical way to relate any Y_k to the inputs $\{X_i\}$, where X_1 passes

through H_{1y_k}, X_2 passes through H_{2y_k}, and so on until X_q passes through H_{qy_k}. Schematically,



Consequences of these definitions will now be developed which provide matrix solutions for general multiple-input/multiple-output models when the number of outputs is the same as the number of inputs. It is assumed that all inverse matrix operations can be performed.

DETERMINATION OF $\mathbf{G}_{xy}$ AND $\mathbf{H}_{xy}$

$$\mathbf{Y} = \mathbf{H}_{xy}'\mathbf{X} \tag{7.136}$$

$$\mathbf{Y}' = (\mathbf{H}_{xy}'\mathbf{X})' = \mathbf{X}'\mathbf{H}_{xy} \tag{7.137}$$

$$\mathbf{X}^*\mathbf{Y}' = \mathbf{X}^*\mathbf{X}'\mathbf{H}_{xy} \tag{7.138}$$

Taking expected values of both sides and multiplying by $(2/T)$ gives

$$\mathbf{G}_{xy} = \mathbf{G}_{xx}\mathbf{H}_{xy} \tag{7.139}$$

This shows how to obtain $\mathbf{G}_{xy}$ from $\mathbf{G}_{xx}$ and $\mathbf{H}_{xy}$. Multiply both sides of Equation (7.139) by $\mathbf{G}_{xx}^{-1}$ to obtain

$$\mathbf{G}_{xx}^{-1}\mathbf{G}_{xy} = \mathbf{G}_{xx}^{-1}(\mathbf{G}_{xx}\mathbf{H}_{xy}) \tag{7.140}$$

where

$$\mathbf{G}_{xx}^{-1} \text{ is the } \textit{inverse} \text{ matrix of } \mathbf{G}_{xx} \tag{7.141}$$

Equation (7.140) is the same as

$$\mathbf{H}_{xy} = \mathbf{G}_{xx}^{-1}\mathbf{G}_{xy} \tag{7.142}$$

This shows how to obtain $\mathbf{H}_{xy}$ from $\mathbf{G}_{xx}$ and $\mathbf{G}_{xy}$. The result of Equation (7.139) is easily pictured as the product of two $q \times q$ matrices $\mathbf{G}_{xx}$ and $\mathbf{H}_{xy}$ to give $\mathbf{G}_{xy}$.

DETERMINATION OF $\mathbf{G}_{yy}$ and $\mathbf{G}_{xx}$

$$\mathbf{Y} = \mathbf{H}'_{xy}\mathbf{X} \tag{7.143}$$

$$\mathbf{Y}' = (\mathbf{H}'_{xy}\mathbf{X})' = \mathbf{X}'\mathbf{H}_{xy} \tag{7.144}$$

$$\mathbf{Y}^* = (\mathbf{H}'_{xy}\mathbf{X})^* = \mathbf{H}'^*_{xy}\mathbf{X}^* \tag{7.145}$$

$$\mathbf{Y}^*\mathbf{Y}' = (\mathbf{H}'^*_{xy}\mathbf{X}^*)(\mathbf{X}'\mathbf{H}_{xy}) \tag{7.146}$$

Taking expected values of both sides and multiplying by $(2/T)$ gives

$$\mathbf{G}_{yy} = \mathbf{H}'^*_{xy}\mathbf{G}_{xx}\mathbf{H}_{xy} \tag{7.147}$$

It follows that

$$\mathbf{G}_{xx} = (\mathbf{H}'^*_{xy})^{-1}\mathbf{G}_{yy}(\mathbf{H}_{xy})^{-1} \tag{7.148}$$

assuming the required inverse matrices exist. The result of Equation (7.147) is easily pictured as the product of three $q \times q$ matrices, $\mathbf{H}'^*_{xy}$, $\mathbf{G}_{xx}$, and $\mathbf{H}_{xy}$, to give $\mathbf{G}_{yy}$. The complex conjugate transpose matrix is as follows.

$$\mathbf{H}'^*_{xy} = \begin{bmatrix} H^*_{1y_1} & H^*_{2y_1} & \cdots & H^*_{qy_1} \\ H^*_{1y_2} & H^*_{2y_2} & \cdots & H^*_{qy_2} \\ \vdots & \vdots & & \vdots \\ H^*_{1y_q} & H^*_{2y_q} & \cdots & H^*_{qy_q} \end{bmatrix} \tag{7.149}$$

Equations (7.139), (7.142), (7.147), and (7.148) are the main matrix formulas to solve multiple-input/multiple-output problems.

Matrix formulas will next be stated for a general multiple-input/single-output model. As before, the input records are not ordered or conditioned in any way.

7.4.2 *Multiple-Input/Single-Output Model*

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_q \end{bmatrix} \qquad \mathbf{H}_{xy} = \begin{bmatrix} H_{1y} \\ H_{2y} \\ \vdots \\ H_{qy} \end{bmatrix} \qquad \mathbf{Y} = Y \tag{7.150}$$

$$Y = \mathbf{H}'_{xy}\mathbf{X} = \begin{bmatrix} H_{1y} & H_{2y} & \cdots & H_{qy} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_q \end{bmatrix} \tag{7.151}$$

$$Y' = Y = \mathbf{X}'\mathbf{H}_{xy} = \begin{bmatrix} X_1 & X_2 & \cdots & X_q \end{bmatrix} \begin{bmatrix} H_{1y} \\ H_{2y} \\ \vdots \\ H_{qy} \end{bmatrix} \tag{7.152}$$

$$\mathbf{X}^* = \begin{bmatrix} X_1^* \\ X_2^* \\ \vdots \\ X_q^* \end{bmatrix} \qquad \mathbf{X}' = \begin{bmatrix} X_1 & X_2 & \cdots & X_q \end{bmatrix} \tag{7.153}$$

The input spectral density matrix is here

$$\mathbf{G}_{xx} = \frac{2}{T}E\{\mathbf{X}^*\mathbf{X}'\} = \begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1q} \\ G_{21} & G_{22} & \cdots & G_{2q} \\ \vdots & \vdots & & \vdots \\ G_{q1} & G_{q2} & \cdots & G_{qq} \end{bmatrix} \tag{7.154}$$

The input/output column vector is

$$\mathbf{G}_{xy} = \frac{2}{T}E\{\mathbf{X}^*\mathbf{Y}\} = \begin{bmatrix} G_{1y} \\ G_{2y} \\ \vdots \\ G_{qy} \end{bmatrix} \tag{7.155}$$

The output spectral density function is

$$G_{yy} = \frac{2}{T}E\{Y^*Y'\} = \frac{2}{T}E\{\mathbf{H}'^*_{xy}\mathbf{X}^*\mathbf{X}'\mathbf{H}_{xy}\} = \mathbf{H}'^*_{xy}\mathbf{G}_{xx}\mathbf{H}_{xy} \tag{7.156}$$

Equation (7.156) is equivalent to writing

$$G_{yy} = \begin{bmatrix} H^*_{1y} & H^*_{2y} & \cdots & H^*_{qy} \end{bmatrix} \begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1q} \\ G_{21} & G_{22} & \cdots & G_{2q} \\ \vdots & \vdots & & \vdots \\ G_{q1} & G_{q2} & \cdots & G_{qq} \end{bmatrix} \begin{bmatrix} H_{1y} \\ H_{2y} \\ \vdots \\ H_{qy} \end{bmatrix} \tag{7.157}$$

Also

$$G_{yy}^{*} = \mathbf{H}_{xy}' \mathbf{G}_{xx}^{*} \mathbf{H}_{xy}^{*} \tag{7.158}$$

$$\mathbf{G}_{xx}^{*} = \mathbf{G}_{xx}' \tag{7.159}$$

For the multiple-input/single-output model,

$$G_{yy} = G_{yy}^{*} = G_{yy}' \tag{7.160}$$

$$\mathbf{G}_{xx}^{*} = (\mathbf{H}_{xy}')^{-1} G_{yy} (\mathbf{H}_{xy}^{*})^{-1} \tag{7.161}$$

$$\mathbf{G}_{xx} = (\mathbf{H}_{xy}'^{*})^{-1} G_{yy} (\mathbf{H}_{xy})^{-1} \tag{7.162}$$

$$\mathbf{G}_{xy} = \mathbf{G}_{xx} \mathbf{H}_{xy} \tag{7.163}$$

Equation (7.163) is equivalent to writing

$$\begin{bmatrix} G_{1y} \\ G_{2y} \\ \vdots \\ G_{qy} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1q} \\ G_{21} & G_{22} & \cdots & G_{2q} \\ \vdots & \vdots & & \vdots \\ G_{q1} & G_{q2} & \cdots & G_{qq} \end{bmatrix} \begin{bmatrix} H_{1y} \\ H_{2y} \\ \vdots \\ H_{qy} \end{bmatrix} \tag{7.164}$$

Note also that

$$\mathbf{H}_{xy} = \mathbf{G}_{xx}^{-1} \mathbf{G}_{xy} \tag{7.165}$$

follows directly from Equation (7.163), where $\mathbf{G}_{xx}^{-1}$ is the inverse matrix of $\mathbf{G}_{xx}$. It is assumed here that the model is well defined and that all inverse matrices exist.

7.4.3 *Model with Output Noise*

Consider now a more realistic multiple-input/single-output model where extraneous uncorrelated output noise can occur. Instead of the ideal Equation (7.151), Y is given by

$$Y = \mathbf{H}_{xy}' \mathbf{X} + N \tag{7.166}$$

where $N = N(f)$ is the Fourier transform of the output noise $n(t)$. In place of Equation (7.156), one now obtains

$$G_{yy} = \mathbf{H}_{xy}'^{*} \mathbf{G}_{xx} \mathbf{H}_{xy} + G_{nn} \tag{7.167}$$

No change occurs in Equations (7.163) so that $\mathbf{G}_{xy}$ with extraneous noise present is still

$$\mathbf{G}_{xy} = \mathbf{G}_{xx} \mathbf{H}_{xy} \tag{7.168}$$

Thus, G_{yy} can be expressed as

$$G_{yy} = \mathbf{H}_{xy}'^{*} \mathbf{G}_{xy} + G_{nn} \tag{7.169}$$

The input spectral density matrix $\mathbf{G}_{xx}$ of Equation (7.154) is a $q \times q$ Hermitian matrix. Define an augmented spectral density matrix of the output $y(t)$ with the inputs $x_i(t)$ by the $(q+1) \times (q+1)$ Hermitian matrix

$$\mathbf{G}_{yxx} = \begin{bmatrix} G_{yy} & G_{y1} & G_{y2} & \cdots & G_{yq} \\ G_{1y} & G_{11} & G_{12} & \cdots & G_{1q} \\ G_{2y} & G_{21} & G_{22} & \cdots & G_{2q} \\ \vdots & \vdots & \vdots & & \vdots \\ G_{qy} & G_{q1} & G_{q2} & \cdots & G_{qq} \end{bmatrix} \tag{7.170}$$

The determinant $|\mathbf{G}_{yxx}|$ of this augmented matrix will now be shown to be zero for all f in an ideal noise-free situation, where $G_{nn} = 0$ in Equation (7.169).

The G_{iy} terms for $i = 1, 2, \ldots, q$ in the first column of $\mathbf{G}_{yxx}$ (below the G_{yy} term) are linear combinations of the G_{ij} terms that appear in the columns of its row. That is, from Equation (7.164),

$$G_{iy} = \sum_{j=1}^{q} H_{jy} G_{ij} \qquad i = 1, 2, \ldots, q \tag{7.171}$$

For the ideal noise-free case, the output spectrum is

$$G_{yy} = \mathbf{H}_{xy}'^{*} \mathbf{G}_{xy} = \mathbf{H}_{xy}' \mathbf{G}_{yx} \tag{7.172}$$

where the last relation occurs because $G_{yy} = G_{yy}^{*}$ and $\mathbf{G}_{xy}^{*} = \mathbf{G}_{yx}$. Equation (7.172) is the same as

$$G_{yy} = \sum_{i=1}^{q} H_{iy} G_{yi} \tag{7.173}$$

Thus, G_{yy} is a linear combination of the G_{yi} terms that appear in the columns of the first row of $\mathbf{G}_{yxx}$. It follows that the matrix $\mathbf{G}_{yxx}$ is such that its entire first column is the result of linear combinations from corresponding terms in the other columns. By a theorem proved in Reference 7.4, the determinant of

this matrix must then be zero, that is

$$|\mathbf{G}_{yxx}| = 0 \quad \text{when } G_{nn} = 0 \tag{7.174}$$

Return to the noise model where G_{yy} is given by Equation (7.169) with $G_{nn} \neq 0$. As noted earlier, no change occurs in Equation (7.163) whether or not extraneous output noise is present. Hence, by using the results derived for the noise-free case, the determinant $|\mathbf{G}_{yxx}|$ of the matrix $\mathbf{G}_{yxx}$ can be computed at any f by the formula

$$|\mathbf{G}_{yxx}| = G_{nn}|\mathbf{G}_{xx}| \tag{7.175}$$

where $|\mathbf{G}_{xx}|$ is the determinant of the matrix G_{xx}. Note that this equation gives $|\mathbf{G}_{yxx}| = 0$ when $G_{nn} = 0$.

From Equation (7.35), it follows that the multiple coherence function is given by the determinant formula

$$\gamma^2_{y:x} = 1 - \left(\frac{|\mathbf{G}_{yxx}|}{G_{yy}|\mathbf{G}_{xx}|} \right) \tag{7.176}$$

As a check on this formula, when $q = 1$, corresponding to a single-input/single-output model,

$$\mathbf{G}_{yxx} = \begin{bmatrix} G_{yy} & G_{yx} \\ G_{xy} & G_{xx} \end{bmatrix} \tag{7.177}$$

Here, the determinants

$$|\mathbf{G}_{yxx}| = G_{xx}G_{yy} - |G_{xy}|^2 \qquad |\mathbf{G}_{xx}| = G_{xx} \tag{7.178}$$

Substitution into Equation (7.176) yields

$$\gamma^2_{y:x} = 1 - \left(\frac{G_{xx}G_{yy} - |G_{xy}|^2}{G_{xx}G_{yy}} \right) = \frac{|G_{xy}|^2}{G_{xx}G_{yy}} \tag{7.179}$$

which is the required ordinary coherence function for this situation.

7.4.4 *Single-Input/Single-Output Model*

$$\mathbf{X} = \mathbf{X}' = X \qquad \mathbf{Y} = \mathbf{Y}' = Y \tag{7.180}$$

$$\mathbf{H}_{xy} = \mathbf{H}'_{xy} = H \tag{7.181}$$

$$Y = HX \tag{7.182}$$

$$G_{xx} = \frac{2}{T}E[X^*X] = G^*_{xx} \tag{7.183}$$

$$G_{yy} = \frac{2}{T}E[Y^*Y] = G^*_{yy} \tag{7.184}$$

$$G_{xy} = \frac{2}{T}E[X^*Y] \qquad G^*_{xy} = G_{yx} \tag{7.185}$$

From Equations (7.182)–(7.185), it follows that

$$G_{xy} = HG_{xx} \tag{7.186}$$

with

$$H = \frac{G_{xy}}{G_{xx}} \tag{7.187}$$

$$G_{yy} = H^*G_{xx}H = |H|^2G_{xx} \tag{7.188}$$

with

$$G_{xx} = \frac{G_{yy}}{|H|^2} \tag{7.189}$$

The relations given above are well known results of Chapter 6 for single-input/single-output problems. They are special cases of previously derived matrix formulas in Sections 7.4.1 and 7.4.2, and show that definitions used there are appropriate for multiple-input/multiple-output problems and multiple-input/single-output problems.

PROBLEMS

7.1 Consider a two-input/single-output system with uncorrelated output noise as shown below.

